Coexisting Exchange Platforms: Limit Order Books and Automated Market Makers

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Abstract

Blockchain-based decentralized exchanges have adopted automated market makers (AMM)—algorithms that aggregate liquidity and automatically set asset prices. This paper analyzes coexisting market structures, a limit-order book (LOB) and an AMM, to investigate their interactions in terms of liquidity. Based on their fundamental design differences, we show that fluctuations in liquidity in the AMM stimulate disproportionate reactions from informed and noise traders, leading to a positive spillover effect on liquidity in the LOB market. Furthermore, unlike models that primarily focus on the AMM, the coexisting environment leads to a hump-shaped reaction of liquidity supply in the AMM to asset volatility.

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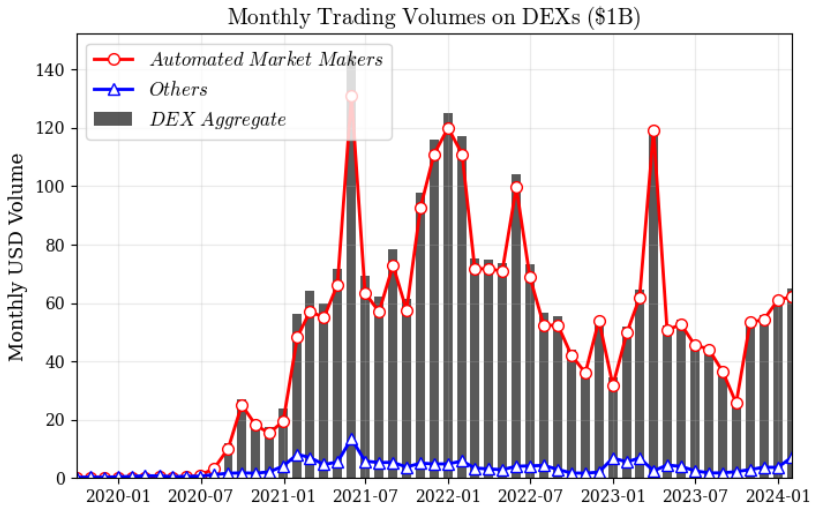
1 Introduction

The modern financial market witnesses substantial fragmentation of trading venues, exemplified by the National Securities Exchanges in the US for stock trading. Nevertheless, they adhere to the same regulatory framework set forth by the SEC and generally employ a single market design—an open electronic limit-order book (LOB). In the crypto market, a similar trend of high fragmentation is observed among platforms, with most centralized exchanges (CEX), such as Binance, utilizing the LOB mechanism. However, decentralized exchanges (DEX) on blockchains, like Uniswap, have embraced a distinctive market structure known as Automated Market Makers (AMM), where a predetermined single-function algorithm dictates asset prices and prevents liquidity providers from setting their own quotes. Despite this restricted strategy space for liquidity provision, AMMs have experienced significant growth in trading share since 2020, as illustrated by Figure 1.

Figure 1: Monthly trading volume on Ethereum-based DEXs



Note: This figure plots the monthly trading volume on DEXs denominated by USD. It includes all DEXs on the Ethereum blockchain through February 2024 and does not cover those on other blockchains, such as Binance Smart Chain. Source: Dune Analytics (duneanalytics.com)

Several studies in the literature have investigated the efficiency of this emerging market structure (e.g., Angeris and Chitra, 2020; Capponi and Jia, 2021; Cao, Hemenway Falk

and Tsoukalas, 2023; Park, 2023) and have compared it with the traditional LOB (Foley, O'Neill and Putniņš, 2023; Malinova and Park, 2024; Lehar and Parlour, 2024). However, most papers analyze AMM and LOB in two independent equilibria and have not explored the interaction between these market structures. Given the significant growth of DEXs and the fact that exchanges are unlikely to coordinate to adopt the same trading mechanism simultaneously, investigating the implications of the coexistence of LOB and AMM is crucial both for academia and practitioners.

In this paper, we analyze the interaction of liquidity between two market structures by considering traders' endogenous platform choice.[1] Our main finding reveals liquidity complementarity between the AMM and the LOB: when AMM liquidity improves, it positively influences LOB liquidity, resulting in a reduction of its bid-ask spread. As discussed below, this liquidity spillover effect represents a direct contrast to the traditional liquidity externality observed among platforms with an identical design. We attribute this result to the fundamental differences in the design of AMM and LOB, which in turn affect how these market structures translate the adverse selection problem into the costs of liquidity incurred by different types of liquidity takers.

We consider a model where liquidity providers (market makers) and takers trade a risky asset on two exchanges: the AMM and the LOB.[2] The trading environment is perfectly competitive, and liquidity takers are either informed or uninformed (noise traders). They choose their trading platforms by comparing the marginal price proposed by the AMM with the bid and ask prices on the LOB, where the bid-ask spread is determined by the adverse selection cost for LOB market makers (e.g., Glosten and Milgrom, 1985).

In contrast to a limit-order market, where participants trade with each other, participants of the AMM trade against *liquidity pools*, i.e., pools of assets reserved on an exchange. For an illustration of its algorithmic pricing, consider a trade between two

[1] Harvey, Ramachandran and Santoro (2021) argue that decentralized finance (DeFi) can mitigate inefficiencies of traditional finance, such as limited access and interoperability. We take these motivations for adopting DeFi as given and focus on the economic consequences of using the new market-making structure.

[2] Hereafter, we use the term the AMM (resp. the LOB) to represent not only a market-making algorithm but also an exchange using it.

digital assets, Ethereum (ETH) and Wrapped Bitcoin (WBTC).[3] Initially, liquidity providers lock these assets into an exchange to create the liquidity pool. Assuming that it contains E and B units of ETH and WBTC, if a trader intends to purchase δ units of WBTC by paying $P\delta$ of ETH, then this order causes a change in the pool's volume from (E, B) to $(E', B') \equiv (E + P\delta, B - \delta)$. The AMM determines the execution price P by requiring that $f(E, B) = f(E', B')$ with pre-determined function f. For example, the Constant Product Market Maker (CPMM), the most commonly used AMM, employs $f(E, B) = EB$, resulting in $P = \frac{E}{B-\delta}$. Liquidity providers withdraw their contributions to the pool upon exiting the market, obtaining changes in the pool size, $E' - E$ and $B' - B$, and trade-execution fees as compensation. In this market, liquidity is determined by the quantity of digital assets that are locked into the platform, (E, B), as this quantity dictates the price impact of a trade.

We find that an increase in AMM liquidity reduces the bid-ask spread and enhances the liquidity of the LOB. This phenomenon stems from the heterogeneous reactions of traders to shifts in AMM liquidity. Additional liquidity injected into the AMM pool diminishes the price impact. Informed traders benefit more from this change compared to noise traders, as they act on the same news and collectively trade a sizable volume. In contrast, due to offsetting random buy and sell orders, each noise trader anticipates a smaller price impact and becomes less sensitive to changes in AMM liquidity. Consequently, informed traders disproportionately migrate away from the LOB to the AMM, thus alleviating the adverse selection problem in the LOB and fostering a positive liquidity spillover effect.

What design difference between these market structures is key to the heterogeneous trader reactions and the liquidity spillover effect? Our argument posits that it stems from their contrasting strategy spaces for liquidity provision and their approaches to compensating market makers for adverse selection.

On the LOB, market makers adjust the price and quantity of their limit orders to countervail adverse selection, and thus severe friction leads to a wider spread and higher price impact (Glosten and Milgrom, 1985; Kyle, 1985). Importantly, the spread is strictly posi-

[3] Wrapped Bitcoin is an ERC-20 token that is pegged to Bitcoin.

tive even for an infinitesimal trade, as liquidity provision is an *ex-ante* commitment, and the price cannot be set contingent on the size of incoming orders (Glosten, 1994). Consequently, both informed traders and (small) noise traders face illiquidity costs, which increase with adverse selection.

In contrast, AMM liquidity providers determine the quantity of liquidity supply but cannot directly control the price, as the AMM algorithm automatically adjusts it by executing incoming swap orders. Although this structure restricts liquidity providers' strategy spaces, the deposited resources are transformed into a *supply schedule* so that the price is automatically and continuously adjusted based on the size of incoming orders. Despite the fact that liquidity provision is an *ex-ante* commitment, the automatic price adjustment essentially empowers liquidity providers to discriminate against relatively larger informed traders, eliminating a positive spread for a small noise trader. Instead, the platform charges a constant transaction fee to liquidity takers and transfers it to liquidity providers on a pro-rata basis, motivating their contributions to liquidity pools (Lehar and Parlour, 2024).

In this environment, liquidity providers can influence the slope of the pricing curve by adjusting the amount of liquidity. Hence, similar to the LOB market, the trading cost for informed traders (i.e., the price impact) increases with adverse selection. Unlike the LOB market, however, small noise traders in the AMM do not pay a positive spread but bear a constant transaction fee to compensate liquidity providers. Since this fee does not increase with adverse selection, noise traders tend to be less responsive to changes in AMM liquidity, resulting in heterogeneous reactions among informed and noise traders.

While the finance literature has established the liquidity externality between platforms of an identical design (e.g., Pagano, 1989), it suggests that liquidity tends to concentrate on one platform, leading to a self-fulfilling equilibrium. Liquidity begets liquidity within a platform, but liquidity among multiple platforms exhibits substitution. Drawing robust predictions from such an equilibrium tends to be challenging, as small changes in parameters may lead to substantially different consequences.

When different market designs coexist for trading the same asset, their heterogeneity in strategy spaces and the reward structure for liquidity provision are key to understand-

ing liquidity interplay. Our model indicates that positive liquidity spillover becomes a robust observation if the liquidity cost imposed on small noise traders exhibits a positive reaction to adverse selection in one platform, while becoming irresponsive in the other platform. In the present financial market, AMM and LOB represent the sole examples of such coexistence, with each holding a significant market share.[4]

Furthermore, we endogenize the liquidity supply to the AMM and conduct comparative statics. Contrary to previous research, our model predicts that the equilibrium AMM liquidity exhibits a hump-shaped response to asset volatility. As suggested by previous studies (e.g., Angeris, Evans and Chitra, 2020; Lehar and Parlour, 2024), higher volatility exacerbates the impermanent loss (i.e., adverse selection) for AMM liquidity providers, discouraging their liquidity supply. At the same time, however, the bid-ask spread on the LOB widens due to more severe adverse selection, inducing traders on the LOB to switch to the AMM. Due to the similar mechanism to the liquidity spillover, we show that migration of noise traders from the LOB to the AMM exceeds that of informed traders, mitigating the adverse selection problem for AMM liquidity providers. This effect encourages liquidity provision to the AMM, resulting in the hump-shaped total reaction of AMM liquidity to asset volatility.

Finally, Section 5 considers several extensions of the baseline model by relaxing assumptions regarding traders' behavior and considering the latest and future versions of AMMs, such as those implemented by Uniswap v3 and v4. In particular, the *concentrated liquidity* algorithm in Uniswap v3 needs careful scrutiny, as it expands the liquidity providers' strategy space by allowing them to select a price range from discrete price ticks, in which their liquidity is actively traded and earns rewards. Based on the pro-rata allocation of rewards, we demonstrate that equilibrium competition among liquidity

[4]The absence of a positive spread resembles the pricing by specialists in hybrid markets (e.g., NYSE) analyzed by, for example, Seppi (1997), Parlour and Seppi (2003), and Bolton, Santos and Scheinkman (2016). However, its underlying mechanism and implication for the LOB bid-ask spread are crucially different: specialists do not need to charge spreads because they decide on the terms of trade *after* observing incoming orders. Since they hold priority to standing limit orders, they "cream skim" profitable retail orders away from limit-order books, thereby widening the spread.

providers does not lead to a positive bid-ask spread,[5] and constant trading fees paid by noise traders still serve as a compensation for adverse selection. Therefore, the key mechanisms behind the liquidity spillover effect survive in this update—the illiquidity cost for informed traders increases with adverse selection, while that for noise traders stays irresponsive—supporting our main argument.

Related literature. Our study contributes to the general understanding of blockchain, cryptocurrency, and DEXs. The literature on this subject is rapidly expanding with comprehensive reviews by, for example, Chen, Cong and Xiao (2020) and Harvey, Ramachandran and Santoro (2021). Numerous papers analyze the blockchain protocol as a new platform for value transfer,[6] but studies prior to the growth of DEXs do not include details on the market structure of the platform.

Although research on the economic impacts of AMMs is in its infancy, several papers have analyzed their asset-pricing implications, including Angeris, Kao, Chiang, Noyes and Chitra (2019), Angeris and Chitra (2020), Angeris, Evans and Chitra (2020), Aoyagi (2020), Evans (2020), Barbon and Ranaldo (2024), and Milionis, Moallemi, Roughgarden and Zhang (2024).[7] In contrast, this paper focuses on the microstructure implication of AMMs, such as liquidity and adverse selection, when they coexist with the traditional LOBs.

Several complementary papers have analyzed the market structure of AMMs. Capponi and Jia (2021) develop a model of liquidity provision via AMMs, investigating the possibility of a liquidity freeze and the impact on other decentralized applications. Park (2023) argue that the AMM pricing rule facilitates inefficient trading, such as sandwich attacks, and may not be aligned with the optimality for liquidity providers. Hasbrouck,

[5] A positive spread in Uniswap v3 corresponds to the absence of liquidity around the current asset's price p_0, i.e., there exist $p_l < p_u$, such that, $p_0 \in [p_l, p_u]$, and no liquidity providers are willing to trade in this range.

[6] For example, see Malinova and Park (2017), Budish (2018), Pagnotta and Buraschi (2018), Schilling and Uhlig (2019), Cong, Li and Wang (2021), Huberman, Leshno and Moallemi (2021), Abadi and Brunnermeier (2022), and Chiu and Koeppl (2022).

[7] Implementational details of decentralized exchanges are provided by, for example, Warren and Bandeali (2017), Zhang, Chen and Park (2018), and Adams, Zinsmeister and Robinson (2020).

Rivera and Saleh (2023) focus on trading fees and show that an exchange can achieve higher trading volumes by imposing higher transaction fees. Cao, Hemenway Falk and Tsoukalas (2023) dig deeper and relate the optimal fee adjustment to asset volatility. **?** provide a model of fragmented liquidity pools and characterize how liquidity is allocated among pools with heterogeneous fees. Capponi, Jia and Yu (2022), in contrast, analyze the information content of trader activity, showing that blockchain fees are informative.

Papers by Foley, O'Neill and Putniņš (2023), Malinova and Park (2024), and Lehar and Parlour (2024) are more closely related to our study, as they compare the LOB and the AMM separately. Lehar and Parlour (2024) build a model of liquidity provision in the AMM and compare its liquidity cost to that of the LOB, thereby characterizing market conditions for the AMM to be a cheaper platform than the LOB. In a similar vein, Malinova and Park (2024) derive the parameter condition that minimizes the AMM cost and calibrate it using data from the US equity market. Foley, O'Neill and Putniņš (2023) attribute a lower trading cost in the AMM to its passive and lower-cost liquidity provision compared to the LOB liquidity, which is mostly provided by sophisticated high-speed traders at sizable liquidity-provision costs.

The contribution of this paper is to provide the first theoretical framework of coexisting AMMs and LOBs, demonstrating how these market structures interact with each other via traders' platform choice. While the DEXs are rapidly growing and have gained a significant share, the centralized LOB exchanges, such as Binance, still dominate the crypto market. Han, Huang and Zhong (2022) empirically attest that traders respond to prices both on CEXs and DEXs, suggesting that the coexistence matters to market quality.

Finally, this paper draws upon a vast body of literature on market microstructure, particularly the seminal works of Glosten and Milgrom (1985) and Kyle (1985), which offer models of market liquidity based on adverse selection. Additionally, Pagano (1989), Chowdhry and Nanda (1991), and Admati and Pfleiderer (1988) provide insights into liquidity externality and its self-fulfilling concentration among platforms or over time. We relate their canonical arguments to the new context of decentralized exchanges, showing that the coexisting LOBs and AMMs exhibit a robust spillover effect of liquidity and do not lead to a self-fulfilling equilibrium with liquidity concentration.

2 Model

2.1 Environment

Consider a trading game in a two-period economy ($t = 0, 1$) with three types of traders: informed traders, noise traders, and market makers (liquidity providers). They trade a single risky asset with an initial common value v_0, and cash is used as the numeraire.[8]

Figure 2: Timeline of the game

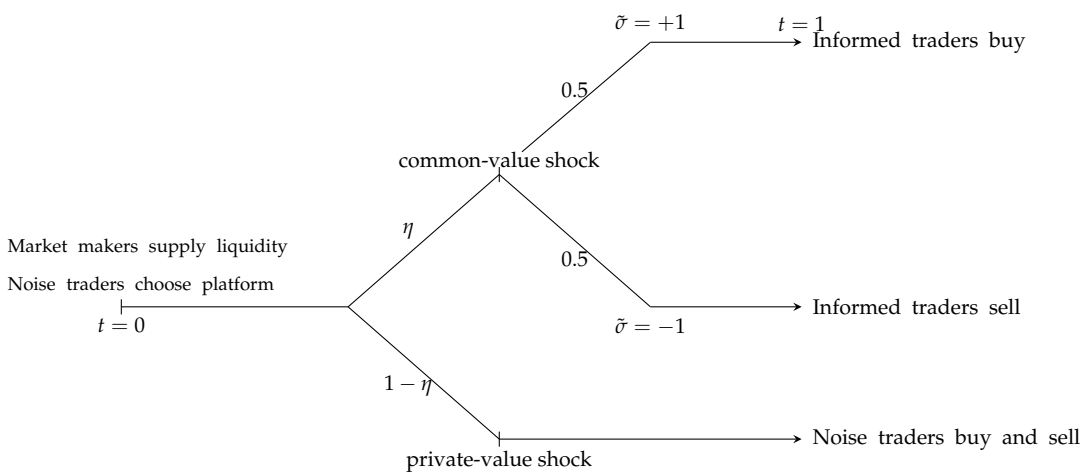


Figure 2 illustrates the timeline of the game, and Table 1 in Appendix A summarizes our notations. Section 5 provides the robustness checks by relaxing assumptions on trader behavior and incorporating the latest update in the AMM implemented by Uniswap.

Events. At $t = 0$, market makers provide liquidity for the subsequent trading round. At $t = 1$, trades occur due to a shock hitting either the common value of the asset or the noise trader's private value of trading. The common-value shock happens with probability $\eta \in (0, 1)$, and the asset's value becomes $\tilde{v} = v_0(1 + \tilde{\sigma})$, where $\tilde{\sigma} = \pm\sigma$ with equal

[8]Throughout the paper, we use the term "asset" to emphasize the generality of the model. Alternatively, we can view the asset as a digital token (e.g., an ERC-20 token) and cash as another risky token or a stable coin, with v_0 representing their relative value. Under this interpretation, the asset price reflects the exchange rate between tokens.

probability.[9] With the complementary probability of $1 - \eta$, the private-value shock triggers noise trading (defined below), and the asset's value remains $\tilde{v} = v_0$. Finally, at the end of $t = 1$, the true value of the asset is realized, and market makers withdraw their liquidity or revoke (reprice) limit orders.

Exchange platforms. Following Zhu (2014), we consider two exchange platforms: a centralized limit-order market (the LOB) and a decentralized automated market (the AMM). The execution price on the AMM is determined by the Constant Function Market Makers (CFMM) specified in Subsection 2.3. Moreover, liquidity takers must pay a transaction fee, $\tau > 0$, to trade in the AMM.[10] It is charged in terms of the asset and accumulated in a separate account (see Lehar and Parlour, 2024).[11] At the end of $t = 1$, fees are distributed to market makers who supplied liquidity to the AMM on a pro-rata basis.

Informed traders. There is a continuum of risk-neutral *informed traders* with a unit measure. If a common-value shock hits the asset at $t = 1$, they immediately observe the realized value of the shock, sequentially arrive at the market, and trade the asset by choosing one of two trading venues. Following market microstructure literature (e.g., Glosten and Milgrom, 1985), each informed trader's asset position is limited to one unit in the baseline model. This restriction can be thought of as a capital constraint, while we relax it in Subsection 5.1 by considering a strategic informed trader trading multiple units of the asset by splitting her order between two platforms.

Noise traders. There is a continuum of noise traders with measure z. The private-value shock triggers their exogenous trading needs, such as those due to hedging motives and

[9] See, for example, Menkveld and Zoican (2017) and Brolley and Zoican (2023) for models with the same structure of shocks. Section 5 generalizes this setting by considering noise traders trading together with informed traders.

[10] τ can be thought of as the *relative* fee in the AMM to that on the LOB, and introducing transaction fees for trading on the LOB does not change our qualitative results.

[11] Our fee structure follows the latest algorithm in Uniswap v3 and v4, as shown in Uniswap Docs. Before these updates, fees are collected and incorporated following a slightly different rule. Appendix B.2 specifies this alternative rule and shows that our results are robust to this change.

margin constraints. With equal probability, each noise trader is forced to buy or sell the asset to the capacity of one unit.[12]

To describe their platform choice, noise traders have heterogeneous private preferences (or aversions) to trading on the AMM or the LOB, characterized by $\gamma \sim U[0, \bar{\gamma}]$ with $\bar{\gamma} > 1$. A type-γ noise trader evaluates the monetary trading cost on the AMM at a discount or with addition of γ (see equation [12] below). Parameter γ summarizes differences between decentralized AMMs and centralized LOBs in a reduced form. For example, $\gamma < 1$ represents traders' aversion to centralized systems due to cyber-security risks and the cost of exploitation by a centralized authority. Similarly, traders with $\gamma > 1$ can be those who dislike decentralized systems due to delays in trade execution (time-consuming blockchain mining processes) or smart contract risks.

We assume that noise traders decide on their trading venues at $t = 0$, i.e., before they enter the market. This could be because they are unsophisticated retail investors, and maintaining multiple accounts on both exchanges or subscribing to a smart order router is costly. Subsection 5.2 allows noise traders to choose platforms contingent on the sign of a private-value shock.

Market makers. Both on the AMM and the LOB, there exist a continuum of uninformed *market makers* (*liquidity providers*). On the LOB, they post a single-unit limit order by quoting competitive bid and ask prices. In the AMM, each market maker locks one unit of the asset in the liquidity pool. We assume that staking liquidity in the AMM incurs a small fixed cost, $\varepsilon > 0$. This cost can be thought of as an opportunity cost of staking resources, gas fees in the Ethereum blockchain, and learning costs to understand the new market mechanism of AMMs. This assumption is to pin down a unique equilibrium and to avoid corner solutions. At the end of $t = 1$, $\tilde{v}$ is realized, and market makers withdraw liquidity from both platforms.[13]

[12] We implicitly assume that noise traders obtain sufficiently large private utility from fulfilling their trading needs so that they participate in the market upon being hit by the shock.

[13] Since our model does not explicitly incorporate Ethereum gas, AMM liquidity providers optimally withdraw liquidity upon learning $\tilde{v}$ to avoid being picked off. Empirically, however, Lehar and Parlour (2024) find infrequent liquidity withdrawal from Uniswap due to the gas payment. In the previous ver-

Solving the model. We denote the measure of informed traders buying (resp. selling) in the AMM as β_{buy} (resp. β_{sell}) and the fraction of noise traders participating in the AMM as α. The following analysis shows that $\beta_{buy} \neq \beta_{sell}$ in general, as the price impact in the AMM is potentially asymmetric between buy and sell orders.

We first analyze the behavior of informed and noise traders at $t = 1$, which is described by their platform choice. Without a loss of generality, we normalize the initial value of the asset to $v_0 = 1$.

2.2 Limit Order Market

The partial equilibrium on the LOB is standard and follows Glosten and Milgrom (1985) and Zhu (2014). We denote the equilibrium bid and ask prices at $t = 1$ as

$$\text{Ask} = 1 + a, \ \text{Bid} = 1 - b.$$

For brevity, a and $-b$ are referred to as the ask and bid prices, and the (effective) bid-ask spread is denoted by $S = a + b$.

Given the venue choice of traders, the expected profits for an LOB market maker from posting a and b on each side of the market are

$$\begin{aligned} V_{ask}^{LOB} &= \frac{1}{2}\left[\eta(1-\beta_{buy})(a-\sigma) + (1-\eta)(1-\alpha)za\right], \\ V_{bid}^{LOB} &= \frac{1}{2}\left[\eta(1-\beta_{sell})(b-\sigma) + (1-\eta)(1-\alpha)zb\right]. \end{aligned}$$

In both equations, the first and the second terms represent a trade with informed and noise traders, which occurs with expected mass $\frac{1}{2}\eta(1-\beta_i)$ and $\frac{1}{2}(1-\eta)(1-\alpha)z$, respectively.

We focus on the perfectly competitive equilibrium in which an LOB market maker breaks even on both sides of the market, leading to the following competitive bid and ask

sion of this paper (available on the authors' webpage), we follow this observation and assume that AMM liquidity providers do not withdraw liquidity until additional cross-market arbitrage completes, showing that this alternative setting does not affect our qualitative results.

prices.[14]

$$a = a(\beta_{buy}, \alpha) = \sigma \frac{(1-\beta_{buy})\eta}{(1-\beta_{buy})\eta + (1-\eta)(1-\alpha)z}, \tag{1}$$

$$b = b(\beta_{sell}, \alpha) = \sigma \frac{(1-\beta_{sell})\eta}{(1-\beta_{sell})\eta + (1-\eta)(1-\alpha)z}. \tag{2}$$

The above prices are potentially asymmetric, as we allow $\beta_{buy} \neq \beta_{sell}$. The bid-ask spread positively reacts to the signal-to-noise ratio (i.e., the measure of informed trading relative to noise trading), because it exacerbates the adverse selection cost for a market maker.

Accordingly, the expected profit for an informed trader who trades on the LOB is given by

$$\pi_I^{LOB}(\tilde{\sigma}) = \begin{cases} \sigma - a(\beta_{buy}, \alpha) & \text{if } \tilde{\sigma} = +\sigma \text{ and buys the asset,} \\ \sigma - b(\beta_{sell}, \alpha) & \text{if } \tilde{\sigma} = -\sigma \text{ and sells the asset.} \end{cases} \tag{3}$$

Similarly, a type-γ noise trader obtains the following *ex-post* profit from trading on the LOB.

$$\pi_{N,l}^{LOB} = \begin{cases} 1 - (1 + a(\beta_{buy}, \alpha)) & \text{if } l = buy, \\ (1 - b(\beta_{sell}, \alpha)) - 1 & \text{if } l = sell, \end{cases} \tag{4}$$

where subscript l indicates the trading direction. Since each noise trader expects to buy and sell with the same probability, her *ex-ante* expected trading cost is identical to the bid-ask spread.

$$\pi_N^{LOB} \equiv \mathbb{E}\left[\pi_{N,l}^{LOB}\right] = -\frac{S(\beta_{buy}, \beta_{sell}, \alpha)}{2}, \tag{5}$$

where the expectation operator is over the direction of a trade ($l = buy, sell$).

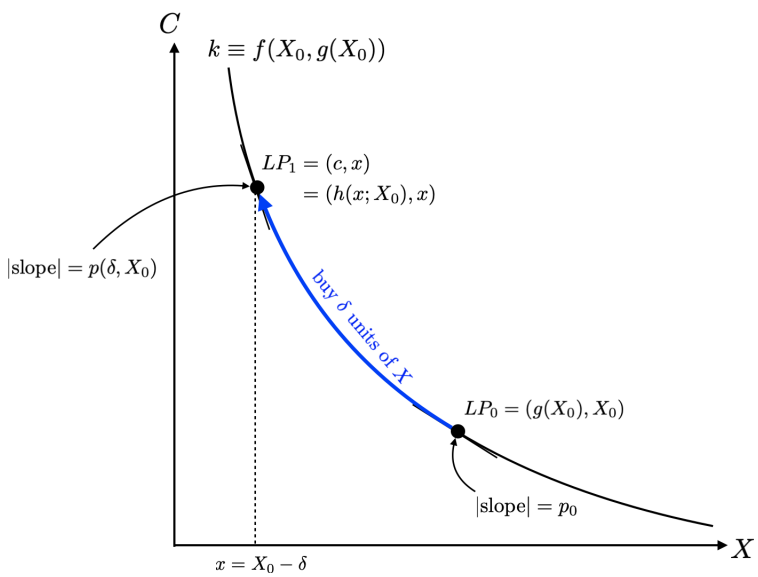
[14]See Glosten and Milgrom (1985) and Zhu (2014). On the LOB, we assume that each one of the continuum of market makers provides a quote for one unit of the asset. Given the measure of the market makers (say, M), each market maker expects to trade with informed and noise traders with probability $\eta(1-\beta_i)/M$ and $(1-\eta)(1-\alpha)z/M$, because all liquidity-taking orders are for one unit, and each limit order is matched with a market order with probability $1/M$. As the competitive market makers break even, M does not appear in the equilibrium equations.

2.3 Automated Market Makers

In what follows, we derive the asset's price based on the AMM algorithm and discuss its crucial difference from the LOB market.

2.3.1 Constant Function Market Makers

Figure 3: Trading with Constant Function Market Makers



Note: This figure illustrates the bonding curve with general $f : \mathbb{R}_+^2 \to \mathbb{R}_+$. LP_0 represents the initial liquidity pools. When X_0 units of the asset is provided, $C_0 = g(X_0)$ is determined so that the slope at L_0 (i.e., the marginal price) has no arbitrage. LP_1 denotes the liquidity pool after δ units of the asset is bought by a trader. The state of the cash pool must be on the bonding curve and is characterized by $c = h(x, X_0)$.

Suppose that the market makers provide X and C units of the aggregate asset and cash to the AMM (Section 4 endogenizes X and C).[15] If an incoming market order trades δ units of the asset ($\delta > 0$ means a buy order) by paying $P\delta$ of cash, the state of the liquidity pools changes as $C \to C' = C + P\delta$, and $X \to X' = X - \delta$. The CFMM is defined by function f that determines the execution price by requiring

$$f(C, X) = f(C + P\delta, X - \delta). \tag{6}$$

[15] Since each market maker with an infinitesimal measure supplies one unit of the asset, X can be seen as the measure of market makers supplying the asset on the AMM.

For example, the CPMM is a special case of (6) with $f(C, X) = CX$. Also, equation (6) draws a curve on the X-C plane, known as the *bonding curve*, as Figure 3 illustrates.

Although we deal with general function f, we impose several regularity conditions to pin down a unique price and to make the bonding curve weakly convex to the origin. The conditions are summarized in Appendix C, and the majority of CFMMs in real-world applications satisfy them (see Xu et al., 2021).

To establish the model with the coexisting exchanges, suppose that no arbitrage exists at the beginning of the game. In the AMM, an infinitesimal trade ($\delta \to 0$) is executed at the marginal price[16]

$$p_0 \equiv \frac{f_x(C, X)}{f_c(C, X)}, \tag{7}$$

where f_j denotes the partial derivative of $f(c, x)$ with respect to $j = c, x$. Therefore, the non-arbitrage condition is expressed as $p_0 = \mathbb{E}[\tilde{v}] = 1$.

Initial cash supply. Given X, condition (7) defines $C = g(X)$ with increasing and differentiable function g, specifying how much cash the market makers must supply to the AMM. It also pins down the CFMM constant $k \equiv f(g(X), X)$ following equation (6). Therefore, liquidity provision is fully characterized by X, and we denote the following results by solely using X instead of denoting $(C, X) = (g(X), X)$.

The bonding curve. In addition, the regularity conditions in Appendix C characterize liquidity pools (c, x) on the bonding curve. Namely, if state (c, x) lies on the bonding curve with the initial asset X, it is expressed as $c = h(x; X)$, where h is a monotonically decreasing convex function. Henceforth, we denote $c = h(x)$ and call it the bonding curve.

By using the above functions, we derive the marginal and average prices for a trade. Figure 3 summarizes notations, and all proofs for the following results are provided in Appendix C.

Lemma 1. *Given the initial asset X and the bonding curve $c = h(x)$, the average price for a trade*

[16] Taking the first-order derivative of equation (6) with respect to δ and setting $\delta \to 0$ yield p_0.

of size $\delta \neq 0$ is given by

$$P(\delta, X) = \frac{1}{\delta} \int_0^{\delta} \frac{f_x(h(X-\tilde{\delta}), X-\tilde{\delta})}{f_c(h(X-\tilde{\delta}), X-\tilde{\delta})} d\tilde{\delta}. \tag{8}$$

Also, after δ units of the asset is bought from the pool, the marginal price for the next incoming trade is given by

$$p(\delta, X) = \frac{f_x(h(X-\delta), X-\delta)}{f_c(h(X-\delta), X-\delta)}. \tag{9}$$

The marginal price function in (9) satisfies the following conditions.[17]

(i) p is increasing in δ;

(ii) p is decreasing in X if, and only if, $\delta > 0$; and

(iii) $\frac{\partial p(\delta,X)}{\partial \delta} > \frac{\partial p(-\delta,X)}{\partial \delta}$ holds for all $\delta > 0$ and $X > 0$.

For a fixed initial liquidity X, conditions (i) and (iii) in Lemma 1 essentially suggest the monotonicity and convexity of the marginal price function with respect to trade size δ.[18] Moreover, condition (ii) implies that ample liquidity lowers the price impact given the size of a trade, thereby making the market deeper. Based on this condition, we refer to the pool size (X) as the measure of liquidity in the AMM.

Example with CPMM. With the CPMM (i.e., $f(C, X) = CX$), the initial marginal price in (7) is given by $p_0 = C/X$, and the non-arbitrage condition requires $C = g(X) = X$ with the constant product being $k \equiv CX = X^2$. Also, if (c, x) is on the bonding curve, it holds that $c = h(x; X) = \frac{X^2}{x}$. Finally, the average and marginal prices are given by

$$P(\delta, X) = \frac{X}{X-\delta} \text{ and } p(\delta, X) = \frac{X^2}{(X-\delta)^2}. \tag{10}$$

[17] *In addition, we can show that (v) $A(\delta, X) \equiv |p(\delta, X) - 1|$ is log-submodular in (δ, X) for $\delta \neq 0$. We use this condition in the formal proof in Appendix C.*

[18] Technically, our results do not require the marginal price to be convex, as point (iii) is a weaker condition than convexity. However, for the expositional reason, the following discussion refers to (iii) as the convexity condition.

It is straightforward to check that Lemma 1 holds because $\frac{\partial p}{\partial \delta} = \frac{2X^2}{(X-\delta)^3} > 0$, $\frac{\partial p}{\partial X} = \frac{-2\delta X}{(X-\delta)^3}$, and $\frac{\partial^2 p}{\partial \delta^2} = \frac{6X}{(X-\delta)^4} > 0$.

2.3.2 Trading Profit in the AMM

When $\beta \in \{\beta_{buy}, -\beta_{sell}\}$ measure of informed traders trade on the AMM, the additional informed trader with an infinitesimal measure obtains the following profit by joining the AMM and trading to her limit of one unit.

$$\pi_I^{AMM}(\tilde{\sigma}) = \begin{cases} 1 + \sigma - \frac{p(\beta_{buy}, X)}{1-\tau} & \text{if } \tilde{\sigma} = +\sigma \\ \frac{p(-\beta_{sell}, X)}{1+\tau} - (1-\sigma) & \text{if } \tilde{\sigma} = -\sigma. \end{cases} \tag{11}$$

Conversely, upon being hit by the private-value shock, a type-γ noise trader expects that random buy and sell orders of other noise traders cancel out, causing no changes in the state of liquidity pools. Therefore, her expected monetary return from (or the cost of) fulfilling her trading needs by trading on the AMM is

$$\pi_{N,l}^{AMM}(\gamma) = \begin{cases} \gamma\left(1 - \frac{p_0}{1-\tau}\right) & \text{if } l = buy, \\ \gamma\left(\frac{p_0}{1+\tau} - 1\right) & \text{if } l = sell. \end{cases} \tag{12}$$

Note that each noise trader expects to pay the marginal price $p_0 = 1$, as she is infinitesimal, and observing her own shock does not affect her inference regarding other noise traders' behavior.

Remarks on the critical design difference. Liquidity provision via the AMM takes the quantity of liquidity as its strategic variable, and liquidity providers cannot choose the price for their resources. Although this is a natural consequence of automatic price adjustments, it represents a stark design contrast from the LOB market, where liquidity providers select both the quantity and price of liquidity. We argue that this design difference results in the distinctive structures of liquidity costs for small (noise) traders among these platforms.

The LOB allows market makers to charge and control the bid-ask spread (i.e., the

cost for a small round-trip trade), rendering the liquidity cost for a small noise trader increasing in adverse selection. The spread is positive even for an infinitesimal trade because liquidity provision is an *ex-ante* commitment of resources, and the first unit on the top of the book faces the highest pick-off risk (Glosten, 1994; Foucault, Pagano and Röell, 2013).

In contrast, although liquidity providers relinquish control of the price, the automatic and continuous price adjustments by the AMM algorithm essentially form deposited liquidity into a *supply schedule* (equations [9] and [10]), discriminating against large orders and eliminating a positive spread. Due to the absence of the spread, the AMM must charge a transaction fee (τ) to small traders to motivate liquidity providers, while this fee does not increase with the magnitude of adverse selection. Therefore, when adverse selection becomes more severe, liquidity costs for non-toxic small traders on the AMM and the LOB exhibit different reactions. In what follows, we show that this difference fosters opposing responses of informed and noise traders to a shock, generating the liquidity spillover effect.

2.4 Platform Choice

Informed traders select their platforms so that the marginal costs of trading on two exchanges become identical.[19]

$$1 + a(\beta_{buy}, \alpha) = \frac{p(\beta_{buy}, X)}{1 - \tau}, \tag{13}$$

$$1 - b(\beta_{sell}, \alpha) = \frac{p(-\beta_{sell}, X)}{1 + \tau}. \tag{14}$$

In both equations, the LHS represents the marginal trading cost on the LOB, while the RHS represents that on the AMM.[20] If the above equations do not balance, marginal in-

[19] Note that the prices do not converge to $\tilde{v} = 1 \pm \sigma$ in the equilibrium. Essentially, this is due to the presence of noise traders, which prevents the prices on the LOB from converging to $\tilde{v}$ except for the extreme case with $(1 - \beta)/(1 - \alpha) \to \infty$.

[20] Although we assume that individual informed traders trade only one unit of the asset, as a group, they incur the price impact cost both in the LOB and the AMM, as the marginal prices on these platforms are

formed traders switch their venue, and β_i adjusts so that (13) and (14) hold.

A type-γ noise trader participates in the AMM if, and only if, the expected monetary return from trading on the AMM exceeds that from trading on the LOB:[21]

$$\frac{1}{2}\sum_{l=buy,sell}\pi_{N,l}^{AMM}(\gamma) \geq \pi_N^{LOB}.$$

Adopting equations (5) and (12), the above inequality reduces to

$$\gamma\bar{\tau}p_0 \leq \frac{S(\beta_{buy},\beta_{sell},\alpha)}{2}, \tag{15}$$

where $\bar{\tau} \equiv \frac{\tau}{1-\tau^2}$ represents the liquidity cost for the noise trader on the AMM. As mentioned earlier, this cost is independent of the degree of adverse selection, while the expected liquidity cost on the LOB, represented by the spread S, increases with adverse selection. By comparing these costs, noise traders with a relatively small γ tend to trade on the AMM.

Adopting uniformly distributed γ, the measure of noise traders using the AMM satisfies the following condition:[22]

$$\alpha = \Pr\left(\gamma < \frac{S(\beta_{buy},\beta_{sell},\alpha)}{2\bar{\tau}p_0}\right) = \frac{S(\beta_{buy},\beta_{sell},\alpha)}{2\bar{\gamma}\bar{\tau}p_0}. \tag{16}$$

3 Liquidity Spillover Effect

We first define and analyze the following equilibrium to explore the impact of AMM liquidity to LOB liquidity.

increasing in $|\beta_j|$. Section 5 and Appendix B.1 make this point more explicit by relaxing the unit-trading assumption of informed trading.

[21] We assume that a noise trader participates in the AMM when she is indifferent.

[22] Both sides of equation (16) are increasing in α, representing the liquidity-begets-liquidity argument. However, equation (16) does not admit multiple solutions regarding α, because we have $S(\beta_{buy},\beta_{sell},0) > 0$ and $S(\beta_{buy},\beta_{sell},1) = 1$. Our model differs from models with multiple equilibria in the literature, because noise traders always participate in the markets due to the sufficiently large private utility from fulfilling their trading needs.

Definition 1. *Given the AMM liquidity (X), the equilibrium is defined by the set of trader measures and prices, $\{(\beta_i)_{i=buy,sell}, \alpha, a, b, p\}$, such that, (i) $(\beta_i)_{i=buy,sell}$ satisfy indifference conditions (13) and (14) for informed traders, (ii) α satisfies the threshold condition (16) for noise traders, (iii) (a, b) are given by the break-even conditions* (1) and (2) *for LOB market makers, and (iv) the AMM marginal price p is given by (9).*

In what follows, we focus on the following parameter region to avoid a market breakdown.

$$\bar{\tau}\bar{\gamma} \geq \sigma. \tag{17}$$

The condition ensures that a positive measure of noise traders use the LOB, and its violation leads the LOB to break down. This condition fails to hold when the AMM fee is sufficiently low (a small $\bar{\tau}$), noise traders' private preferences are strongly biased toward trading on the AMM (a small $\bar{\gamma}$), or asset volatility is extremely high (a large σ).[23] These situations exogenously make the AMM a strictly dominant platform, thereby taking all noise traders away from the LOB.

Noise traders. Applying the spread equations in (1) and (2), the platform choice of noise traders in (16) is characterized by

$$\bar{\tau}\bar{\gamma}\alpha = \frac{\sigma}{2} \sum_{i=buy,sell} \frac{(1-\beta_i)\eta}{(1-\beta_i)\eta + (1-\eta)(1-\alpha)z}. \tag{18}$$

Lemma 2. *(i) Equation (18) yields a unique measure of AMM noise traders as a function of $(\beta_{buy}, \beta_{sell})$ denoted as $\alpha^* = \alpha^*(\beta_{buy}, \beta_{sell})$.*
(ii) α^* is monotonically decreasing in β_{buy} and β_{sell}.

When more informed traders trade in the AMM (β_i increases), the adverse selection cost on the LOB decreases, leading to a narrower bid-ask spread. Since the trading cost for noise traders in the AMM is unaffected by β_i, the LOB becomes relatively cheaper and more attractive for noise traders, thereby reducing α^*.

[23] Given X, the marginal price of the asset in the AMM does not directly depend on σ, while it directly expands the bid-ask spread on the LOB.

Informed traders. Analogously, the indifference conditions for informed buyers and sellers in equations (13) and (14) are rewritten as

$$\frac{p(\beta_{buy}, X)}{1-\tau} = 1 + \sigma \frac{(1-\beta_{buy})\eta}{(1-\beta_{buy})\eta + (1-\eta)(1-\alpha)z}, \tag{19}$$

$$\frac{p(-\beta_{sell}, X)}{1+\tau} = 1 - \sigma \frac{(1-\beta_{sell})\eta}{(1-\beta_{sell})\eta + (1-\eta)(1-\alpha)z}. \tag{20}$$

Lemma 3. *(i) Equations (19) and (20) yield unique measures of informed buyers and sellers on the AMM as functions of (α, X), which we denote as $\beta_i^* = \beta_i^*(\alpha, X)$ for $i \in \{buy, sell\}$.*
(ii) With X being fixed, β_i^* is monotonically increasing in α.

Point (ii) suggests that more informed traders are willing to use the AMM when the measure of AMM noise traders increases. This is because the migration of noise traders from the LOB to the AMM exacerbates adverse selection on the LOB. The bid-ask spread widens, and the LOB becomes a relatively expensive platform for informed traders.

3.1 Exploring Equilibrium and Liquidity Spillover

By using the results in Lemmas 2 and 3, we solve for the equilibrium measures of AMM traders, their reactions to changes in AMM liquidity, and the implication for LOB liquidity.

Proposition 1. *In the equilibrium, there is a unique set of $(\alpha^*(X), \beta_{buy}^*(X), \beta_{sell}^*(X))$ that solves equations (18)-(20), and the solutions are stable. In this equilibrium;*
(i) The average measure of informed traders in the AMM, $\frac{\beta_{buy}^* + \beta_{sell}^*}{2}$, is increasing in X;
(ii) The measure of the noise traders in the AMM, α^*, is decreasing in X; and
(iii) The bid-ask spread on the LOB is decreasing in X.

Proposition 1 demonstrates that a unique and stable equilibrium exists in terms of traders' platform choice, and exogenous variations in AMM liquidity affect LOB liquidity. Point (i) suggests that informed traders are inclined to participate in the AMM when it becomes more liquid, while point (ii) indicates that noise traders exhibit the opposing reaction by switching from the AMM to the LOB.

The contrasting reactions of informed and noise traders arise from the distinctive responses of liquidity costs on the AMM to changes in its liquidity, as highlighted by the remark in Subsection 2.3.2. Although individual informed traders trade a single unit, they collectively incur the price impact, as they generate a shift in the AMM liquidity pool by trading on the same news. Hence, an increase in the liquidity pool attracts more informed traders by mitigating the price impact (Lemma 1). In turn, more active informed traders on the AMM alleviate adverse selection for the LOB market makers, leading to a decline in the bid-ask spread.

On the contrary, changes in AMM liquidity do not directly impact the choice of noise traders, as they expect to consume only a limited amount of liquidity as a group. Instead, each small noise trader pays the transaction fee (τ) to compensate market makers, whereas this cost is not responsive to adverse selection. Therefore, with a narrower bid-ask spread on the LOB, caused by the migration of informed traders, noise traders tend to move towards the LOB, thereby reducing the spread even more.

The liquidity spillover effect represents a distinctive feature of the coexistence of the LOB and the AMM, contrasting with non-trivial equilibrium in cases of coexisting platforms with identical designs. For instance, in the latter scenario, Pagano (1989) and Chowdhry and Nanda (1991) argue that the equilibrium is not robust (unstable) due to the self-fulfilling nature of liquidity concentration. Namely, if a trader anticipates that other traders use a particular platform, she also trades there, leading to a concentration of liquidity on that platform. In such a context, liquidity among platforms exhibits strategic substitution. A similar observation arises when traders choose the timing of trades within a single platform, as proposed by Admati and Pfleiderer (1988).[24]

In contexts where various market designs coexist, the transmission of liquidity variations across platforms depends on the relative responses of informed and noise traders (i.e., the signal-to-noise ratio) to changes in liquidity, which, in turn, are influenced by changes in their liquidity costs. Proposition 1 indicates that the AMM and the LOB rep-

[24]These models indicate that the coexistence of identical platforms is sustained only in a knife-edge case. Therefore, they introduce different parameters for platforms (e.g., fee levels) or heterogeneous preferences for noise traders to support a coexisting equilibrium.

resent a prominent pair of market designs, each with substantial market shares in reality, where one of them maintains the price of liquidity for small traders relatively insensitive to changes in adverse selection, thus generating a robust liquidity spillover effect on the other market. This result is attributed to their market design differences in terms of liquidity providers' strategy spaces and resulting liquidity costs for informed and noise traders.

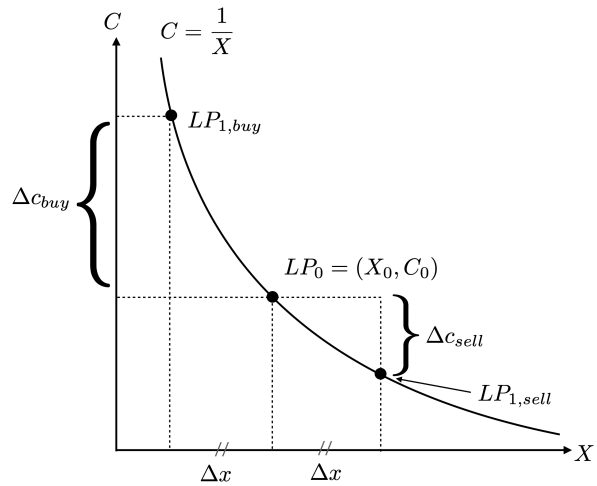
3.2 Asymmetric Price Impact

Another unique implication of the AMM is the asymmetric price impact of buy and sell orders and its influence on bid and ask prices on the LOB.

Lemma 4. *(i)* $\beta^*_{buy} < \beta^*_{sell}$ *for all* $\alpha \in (0, 1)$ *and* X.
(ii) The midpoint quote on the LOB is biased and is higher than the expected value of the asset, i.e., $\frac{Ask+Bid}{2} > 1 = \mathbb{E}[\tilde{v}]$.

To understand intuition, consider Figure 4 with the CPMM as an example. A buy order with size $\delta = \Delta x > 0$ moves the initial liquidity pool LP_0 upward to $LP_{1,buy}$, while a sell order of the same size moves LP_0 downward to $LP_{1,sell}$, both along the bonding curve $C = X^{-1}$. Since the curve is convex, the buy order requires a larger adjustment along the y-axis than the sell order, i.e., $\Delta c_{buy} > \Delta c_{sell}$.

Figure 4: Asymmetric price impact



Note: This figure describes the price impact of buy and sell orders of the same size, Δx, along the CPMM curve.

In general, adding (selling) the asset to the pools bears a smaller price impact than extracting (buying) it due to the convexity of the AMM pricing curve. Hence, an innovation in the asset's value induces disproportional reactions between informed buyers and sellers, resulting in $\beta^*_{buy} < \beta^*_{sell}$. This inequality implies that informed sellers are more likely to use the AMM than informed buyers.

Moreover, this result leads to the asymmetric informativeness (or toxicity) of buy and sell orders on the LOB: a buy order is more likely to be placed by an informed trader than a sell order. As a result, the bid and ask prices are asymmetrically distributed around the expected value of the asset, and the mid-point quote is biased above the *ex-ante* efficient price.

4 Endogenous AMM Liquidity

This section considers liquidity provision to the AMM and analyzes how two market structures jointly respond to shocks in the equilibrium.

4.1 Profits of AMM Liquidity Providers

In order to determine the profit function of AMM liquidity providers, we assume they unwind their asset positions at price $\tilde{v}$ (e.g., by trading on the LOB) upon withdrawing liquidity from the AMM.

In the end of $t = 1$, AMM liquidity providers obtain the following amounts of the asset and cash (incorporating the fee income).

$$(X', C') = \begin{cases} (X - (1-\tau)\beta_{buy}, g(X) + P(\beta_{buy})\beta_{buy}) & \text{if } \tilde{\sigma} = +\sigma, \\ (X + (1+\tau)\beta_{sell}, g(X) - P(\beta_{sell})\beta_{sell}) & \text{if } \tilde{\sigma} = -\sigma, \\ (X, g(X)) & \text{if private-value shock occurs.} \end{cases} \tag{21}$$

The first two lines show the liquidity positions after informed trading, where the changes in cash involve the total payment or extraction by liquidity takers represented by $P(\beta)\beta = \int_0^\beta p(\delta)d\delta$. The last line captures the case of noise trading. By the law of large numbers (LLN), they do not change the state of the liquidity pool, and the marginal price on the AMM stays $p_0 = 1 = \tilde{v}$.

Liquidity provision must follow the AMM algorithm specified in Subsection 2.3. Namely, by staking one unit of the asset, each liquidity provider also commits c units of cash to the cash pool so that the aggregate pool satisfies $C = g(X)$. As a reward for staking liquidity, each liquidity provider receives $w \equiv \frac{1+c}{X+g(X)}$ shares of the end-of-period liquidity pool in equation (21).[25]

Firstly, suppose that the trigger event is the common-value shock. Each liquidity

[25]The value of c depends on the assumption regarding the timing of liquidity provision. For example, c for the first-arriving liquidity provider is different from that for the second liquidity provider due to the change in the pool state. To avoid the complication arising from the arrival order, we assume that all liquidity providers have the same speed technology and simultaneously decide on their supply, where the liquidity is locked into the pool following a Poisson arrival process with the same intensity (i.e., the expected arrival timings for individuals follow a uniform distribution).

provider expects to earn the following net profit from this event:

$$\begin{aligned} V_{IT}^{AMM}(X) =& \frac{w}{2}\left[g(X) + P(\beta_{buy}, X)\beta_{buy} + (1+\sigma)(X - (1-\tau)\beta_{buy})\right. \\ & \left. + g(X) - P(-\beta_{sell}, X)\beta_{sell} + (1-\sigma)(X + (1+\tau)\beta_{sell})\right] - (c+1) \\ =& \frac{w}{2}\left\{\left[P(\beta_{buy}, X) - (1+\sigma)(1-\tau)\right]\beta_{buy} + \left[(1-\sigma)(1+\tau) - P(-\beta_{sell}, X)\right]\beta_{sell}\right\}. \end{aligned} \tag{22}$$

The first and the second lines represent the post-trade net values of the liquidity pool when $\tilde{v} = 1 \pm \sigma$, respectively, while the last line expresses them in terms of the profit margin of incoming trades, β_j. The following results hold for V_{IT}^{AMM}.

Lemma 5. *(i) The liquidity provider's expected net profit from executing informed trading is negative, i.e., $V_{IT}^{AMM}(X) < 0$ for all X.*
(ii) $V_{IT}^{AMM}(X)$ is monotonically decreasing in X.

In DeFi literature, the loss incurred due to the informed trading is referred to as the *impermanent loss* (e.g., Angeris and Chitra, 2020). When a trade occurs due to information about $\tilde{v}$, the liquidity pools consistently lose a more valuable asset and acquire a less valuable asset, incurring a decline in the pool value. This observation accentuates the similarity between AMMs and LOBs, where informed trading is toxic due to its adverse selection cost on market makers (Glosten and Milgrom, 1985; Kyle, 1985). Additionally, the impermanent loss increases with the size of the liquidity pool (X), as it attracts more informed traders to the AMM (Proposition 1).

Secondly, when the private-value shock triggers trades at $t = 1$, liquidity providers earn positive profits, as noise traders do not change the liquidity pools on average but pay the trading fee. By incorporating the absolute volume of noise trading, $z\alpha$, the expected net profit is

$$V_{NT}^{AMM} = w\tau z\alpha. \tag{23}$$

Lemma 6. *(i) The liquidity provider's expected net profit from trading with noise traders is positive, i.e., $V_{NT}^{AMM} > 0$ for all X.*
(ii) V_{NT}^{AMM} is monotonically decreasing in X.

As the adverse selection cost is absent, liquidity providers obtain a strictly positive return from trading with noise traders. Since the mass of noise traders (α) is decreasing in the AMM liquidity (Proposition 1), V_{NT}^{AMM} shrinks as the pool becomes larger.

4.2 Liquidity Provision to the Automated Market

The expected profit from providing liquidity to the AMM is the combination of impermanent loss and fee income from noise trading, net of the exogenous opportunity cost $\varepsilon > 0$.

$$V_M^{AMM}(X) = \eta V_{IT}^{AMM}(X) + (1-\eta) V_{NT}^{AMM}(X) - \varepsilon. \tag{24}$$

Note that each liquidity provider is infinitesimal and does not incorporate the impact of her liquidity provision on the equilibrium variables including aggregate X. Since we assume perfect competition in the market-making sector, the equilibrium size of the liquidity pool, denoted as X^*, is determined by the break-even condition, $V_M^{AMM}(X^*) = 0$.

Proposition 2. *$V_M^{AMM}(X)$ is monotonically decreasing in X, and the break-even condition admits a unique and stable solution X^*.*

The equilibrium behavior of informed and noise traders in Proposition 1 provides the intuition behind the above result. As the liquidity pool expands, the AMM attracts more informed traders, while noise traders migrate away from the AMM to the LOB. Therefore, an increase in X is associated with a more severe adverse selection cost and a decline in the fee income from noise trading. The equilibrium liquidity provision is determined by X^* that balances these two competing effects. Since V_M^{AMM} crosses the horizontal line from above, X^* is stable and robust to a perturbation in parameter values.

Once again, the stability of X^* hinges on the opposing reactions of informed and noise traders to changes in AMM liquidity. If all traders react in the same direction, as in the case of coexisting LOBs, whether it improves individual liquidity providers' profit depends on their relative reactions, making it challenging to pin down a robust prediction. This model highlights that the AMM operating in parallel with the LOB does not suffer

from this issue.

4.3 Comparative Statics: Hump-shaped Reaction of AMM Liquidity

This subsection analyzes the break-even condition of equation (24) by exogenously varying asset volatility, σ, to comprehend its impact on the AMM liquidity, X^*. Understanding its impact is of first-order importance in practical liquidity provision to derive strategies for managing risk and optimizing returns in AMM-based protocols. The literature has extensively argued that high volatility can lead to impermanent loss, causing a reduction in overall liquidity and potential loss for liquidity providers. However, our model, incorporating coexisting platforms, leads to a different observation.

To facilitate the numerical analyses, we consider the CPMM algorithm, $f(C, X) = CX$, as the leading example of AMMs in the real market. Figures 5 and 6 illustrate the impact of σ on equilibrium variables with different η (the frequency of the common-value shock) and z (the aggregate measure of noise trading), demonstrating the hump-shaped reaction of the AMM liquidity supply to asset volatility in Panels A.

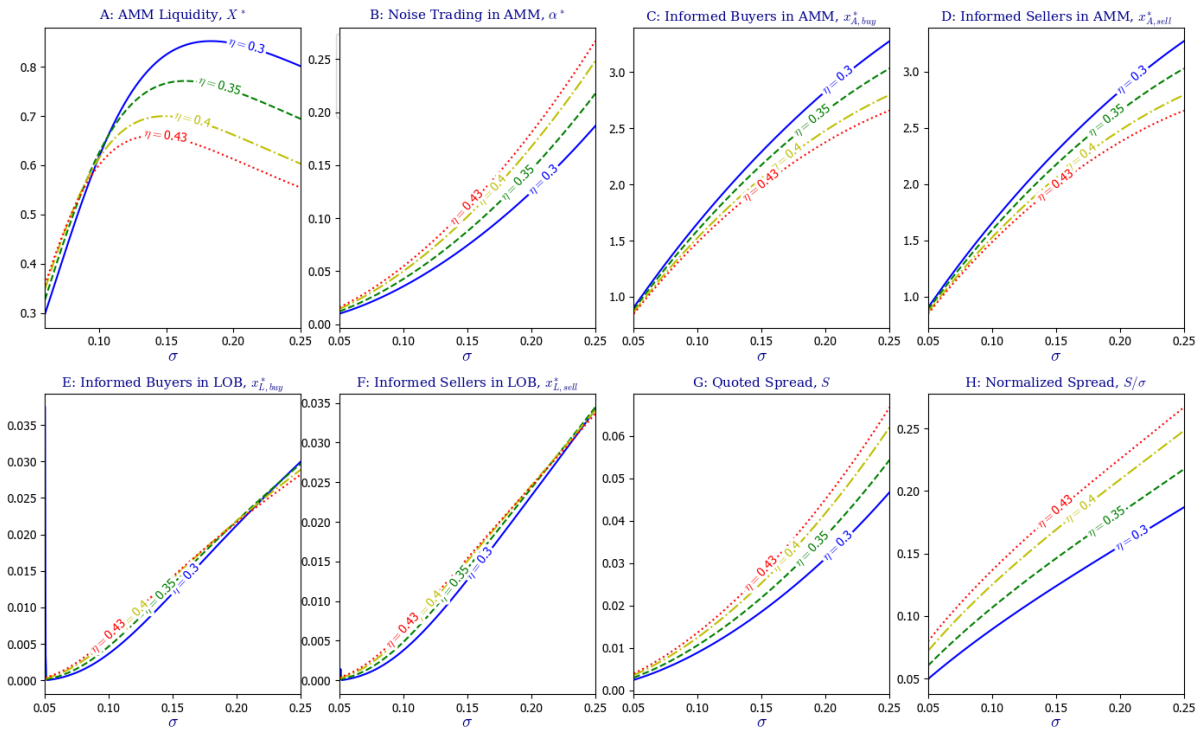
The hump-shaped reaction is in stark contrast to existing studies of AMMs (e.g., Angeris et al., 2019; Lehar and Parlour, 2024), which suggest that the amount of AMM liquidity should globally decrease with asset volatility due to larger impermanent losses. This effect is present in our model; the figures suggest that β_j increases with σ, leading to a larger loss in liquidity provider profits, as expressed by

$$\begin{aligned}\frac{2}{w}\frac{\partial V_{IT}^{AMM}(X)}{\partial \sigma} &= -(1-\tau)\beta_{buy} - (1+\tau)\beta_{sell} \\ &\quad + \frac{1}{1-\tau}\underbrace{\left(\frac{p(\beta_{buy})}{1-\tau} - (1+\sigma)\right)}_{<0}\frac{\partial \beta_{buy}}{\partial \sigma} + \frac{1}{1+\tau}\underbrace{\left((1-\sigma) - \frac{p(-\beta_{sell})}{1+\tau}\right)}_{<0}\frac{\partial \beta_{sell}}{\partial \sigma} \\ &< 0.\end{aligned} \tag{25}$$

Similar to the classic adverse selection argument, higher volatility not only amplifies the cost associated with informed trading (the first line of [25]), but also escalates that intensity itself (the second line of [25]). Since equilibrium liquidity provision is determined by balancing the adverse selection cost and the fee income from noise trading, the above

mechanism leads to a reduction in X^*.

Figure 5: Equilibrium Impact of Asset Volatility with Different η



Note: This figure is illustrated by using $z = 0.5, \bar{\gamma} = 30.0, \tau = 0.03, \varepsilon = 0.025$ and focusing on the parameter region that condition (17) holds. In the parameter regions with no lines, we do not have equilibrium. The informed trading in Panel C plots the average measure of informed traders, $0.5(\beta_{buy} + \beta_{sell})$. The noremalized spread in panel F is S/σ.

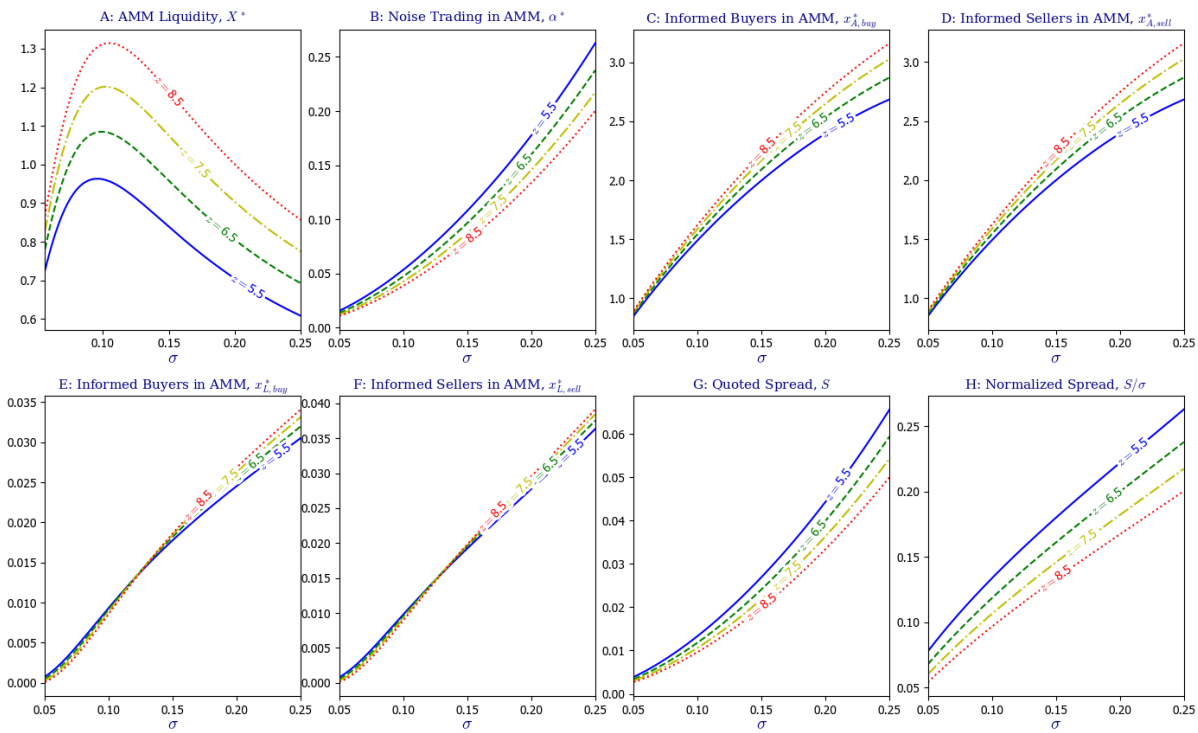
However, the endogenous interaction of liquidity in two market structures generates an opposing effect. As indicated by equations (1) and (2), heightened asset volatility amplifies adverse selection and widens the bid-ask spread on the LOB, making both informed and noise traders more inclined to migrate from the LOB to the AMM (α and β_i increase). Their reactions are summarized by Panels B and C.

Importantly, informed traders are less responsive to changes in the bid-ask spread than noise traders. Upon moving to the AMM, informed traders collectively cause a considerable shift in AMM liquidity pools and incur a relatively large marginal cost, while noise traders do not experience this impact due to their offsetting random trading. Consequently, the migration of noise traders surpasses that of informed traders, mitigating adverse selection for AMM liquidity providers and promoting liquidity supply to the

AMM.

Due to these two competing effects, equilibrium AMM liquidity exhibits a hump-shaped reaction to asset volatility. As Subsection 5.2 and Appendix B demonstrate, this result is robust to relaxing assumptions regarding the measure of noise traders and making them incur a strictly positive price impact as a group.

Figure 6: Equilibrium Impact of Asset Volatility with Different z



FIGURE

Note: This figure is illustrated by using $\eta = 0.7, \bar{\gamma} = 30.0, \tau = 0.03, \varepsilon = 0.025$ and focusing on the parameter region that condition (17) holds. In the parameter regions with no lines, we do not have equilibrium. The informed trading in Panel C plots the average measure of informed traders, $0.5(\beta_{buy} + \beta_{sell})$. The noremalized spread in panel F is S/σ.

Note that the positive reaction of X^* arises from the LOB bid-ask spread, while the opposing effect emanates from the price function of the AMM. The former impact varies linearly with σ (i.e., the effective spread $= \sigma\times$ the normalized spread), while the latter effect manifests through the AMMs' convex pricing function, p. Therefore, the negative

impact of σ on X^* becomes comparatively stronger and tends to dominate the positive effect as σ increases, generating the hump-shaped reaction of X^*.

Also, Figure 5 suggests that the hump-shaped reaction tends to arise when η is large. *Ceteris paribus*, a large η means a high signal-to-noise ratio and a wide normalized bid-ask spread. Hence, the effective spread is more responsive to σ (panel E), and the migration effect of noise traders on liquidity supply tends to be stronger.[26]

5 Robustness Checks and Discussions

This section establishes the robustness of our main results by relaxing assumptions in the baseline model. We also incorporate the latest updates in the AMM. The details of the generalized models and proofs are provided in Appendix B.

5.1 When Informed Trader Trades Multiple Units

The baseline model assumes that each informed trader is small and trades up to one unit of the asset, limiting their price impact costs. This assumption could be restrictive and might introduce a slight bias in comparing the AMM and the LOB; informed traders expect to trade at the best bid and ask prices on the LOB, while they incur the price impact in the AMM as a group. To address this concern, this subsection analyzes a single informed trader trading multiple units of the asset by incorporating the increasing trading costs in both platforms.

5.1.1 Environment

Consider a single informed trader. According to $\tilde{v} = 1 \pm \sigma$, she trades $(x_j)_{j=buy,sell}$ units of the asset in total by splitting her order between the AMM and the LOB, denoted as $x_{A,j}$

[26]The measure of noise trading, z, has a similar but smaller impact on the migration effect, and even a small z leads to the hump-shaped reaction in Figure 6. This is because increases in z affect the spread only via the measure of noise trading and have no impact on informed trading intensity.

and $x_{L,j} = x_j - x_{A,j}$ for $j \in \{buy, sell\}$, respectively.[27] We assume that the informed trader faces a quadratic adjustment cost for the total trading size, $\frac{c}{2}x_j^2$, which could be justified by risk aversion or inventory costs. The structure of the AMM is the same as the baseline model.

On the LOB, in contrast, the first z units of limit orders are available at the competitive bid and ask prices, but these prices are increasing in $x_{L,j}$ reflecting adverse selection. The limit orders beyond z units are offered at the ask and bid prices of $1+\sigma$ and $1-\sigma$, respectively. This price structure is justified by the competitive liquidity provision: market makers know that orders of size $x > z$ must be from the informed trader. In summary, the LOB prices are given by

$$\begin{cases} 1 + \tilde{\sigma} s(x, \alpha) & \text{for } x \leq z, \\ 1 + \tilde{\sigma} & \text{for } x > z. \end{cases}$$

where $\tilde{\sigma} = \pm\sigma$, and the spread component for $x \leq z$ is determined by the break-even condition for market makers:

$$s(x, \alpha) \equiv \frac{x\eta}{x\eta + (1-\eta)(1-\alpha)z}. \tag{26}$$

In what follows, we characterize the equilibrium with $\tilde{v} = 1+\sigma$, while the symmetric argument applies to the case with $\tilde{v} = 1-\sigma$.[28]

Focusing on $x_{L,buy} \leq z$,[29] denote the total trading cost to buy $x_{L,buy}$ units of the asset on the LOB as

$$A(x_{L,buy}) \equiv x_{L,buy}\left(1 + \sigma s(x_{L,buy}, \alpha)\right),$$

[27] As in the baseline model, orders are asymmetric ($x_{buy} \neq x_{sell}$) due to the convexity of the marginal price in the AMM.

[28] Appendix B.1 provides full characterizations and proofs.

[29] As the LOB prices become $\tilde{v}$, the informed trader never trades $x > z$ on the LOB.

with the marginal cost being

$$ask(x_{L,buy}) \equiv \frac{dA(x_{L,buy})}{dx_{L,buy}} = 1 + \sigma s(x_{L,buy}, \alpha) \left(2 - s(x_{L,buy}, \alpha)\right). \tag{27}$$

Note that $1 - s$ in the second term of (27) captures the price impact (or the spread impact). Also, it holds that $\frac{\partial ask}{\partial x_{L,buy}} > 0$, meaning that the trader faces an increasing and convex trading cost on the LOB.

5.1.2 Optimal Informed Trading

The informed trader's expected profit is expressed as

$$V_{buy} = (1+\sigma)x_{buy} - \frac{P(x_{A,buy}, X)}{1-\tau} x_{A,buy} - A(x_{L,buy}) - \frac{c}{2} x_{buy}^2$$

with $x_{buy} = x_{A,buy} + x_{L,buy}$, leading to the following FOCs.

$$\begin{aligned} (x_{buy}) &: 1 + \sigma - ask(x_{L,buy}) - c x_{buy} = 0, \\ (x_{A,buy}) &: -\frac{p(x_{A,buy})}{1-\tau} + ask(x_{L,buy}) = 0. \end{aligned} \tag{28}$$

The first condition in (28) determines the total trading quantity, x_{buy}, given the order-splitting strategy. In contrast, the second condition implies no cross-market arbitrage, meaning that the trader splits x_j between two platforms so that the additional infinitesimal re-allocation cannot generate profits.

In what follows, we consider the following parameter condition to ensure $x_{L,j} < z$, thereby ruling out a trivial equilibrium where the LOB prices become $1 \pm \sigma$.[30]

$$cz > \sigma(1-\eta)^2. \tag{29}$$

Intuitively, the above condition makes it costly to trade an excessively large quantity by exogenously widening the spread (z, σ, η) or by increasing the marginal cost of trading

[30] If $x_{L,j}^* = z$, changes in $x_{A,j}$ due to fluctuations in X does not affect $x_{L,j}$, and the liquidity spillover effect does not arise.

(c).

Lemma 7. *Given α and X, there is a unique solution to the FOCs in (28).*
(i) For $j \in \{buy, sell\}$, $x_{A,j}$ and x_j (resp. $x_{L,j}$) are decreasing (resp. increasing) in α.
(ii) Comparing buy and sell orders, following inequalities hold: $x_{sell} > x_{buy}$, $x_{A,sell} > x_{A,buy}$, and $x_{L,buy} > x_{L,sell}$.

The reaction of informed trading to changes in noise trading is identical to the baseline model. Also, the asymmetric price impact on the AMM results in biased buy and sell orders in both platforms, leading to the asymmetric informativeness of order flows and the biased mid-point quote on the LOB.

For completeness, the following result applies to noise traders, owing to the same structure as the baseline model.

Lemma 8. *Given $x \equiv (x_{A,j}, x_{L,j})_{j=buy,sell}$, the mass of noise traders in the AMM is determined by the unique solution to the following condition:*

$$\alpha = \frac{\sum_{j=buy,sell} s(x_{L,j}, \alpha)}{2\bar{\gamma}\bar{\tau}}. \tag{30}$$

The solution, $\alpha(x_{L,buy}, x_{L,sell})$, is monotonically increasing in $x_{L,j}$.

5.1.3 Exploring Equilibrium and Liquidity Spillover Effect

The following proposition demonstrates the robustness of the liquidity spillover effect to this extension:

Proposition 3. *Given X, there exists a unique equilibrium characterized by the trading volumes of the informed trader, denoted as $x(X) = (x_{A,j}(X), x_{L,j}(X))_{j=buy,sell}$, and the mass of noise traders in the AMM, represented by $\alpha(X)$. In this equilibrium, the bid-ask spread on the LOB decreases as X increases.*

Moreover, the comparative statics of Proposition 3 regarding asset volatility are presented in Figures 8 and 9 in Appendix D, which support a hump-shaped reaction of the AMM liquidity supply to variations in σ.

In summary, we confirm the robustness of the main results in the baseline model. Whether or not the informed trader takes the increasing and convex trading cost in the LOB into account, informed and noise traders still demonstrate opposing reactions to changes in AMM liquidity. The primary driver of these results lies in the AMM market design, which renders liquidity costs for noise traders independent of adverse selection. This fundamental structure remains unaffected even if we generalize the informed trader's trading strategy and the LOB price schedule, provided it charges a spread for a small trader that increases with adverse selection.

5.2 Generalizing Noise Traders

In our baseline model, the expected price impact for noise traders on the AMM is minimized due to the assumptions that (i) they buy and sell with the same expected measure and (ii) do not trade when the common-value shock occurs with probability η (i.e., they do not trade together with informed traders).[31] In what follows, we relax these assumptions.

Suppose that noise traders are always active and engage in trading regardless of the type of the trigger shock. We also allow them to decide on the trading venue *after* observing their shocks (i.e., buy or sell), leading to asymmetric measures of buy and sell noise traders in the AMM, denoted as α_{buy} and α_{sell}, respectively.

5.2.1 LOB Trading Costs and Platform Choice by Informed Traders

The generalization leads to the following expected profits for the LOB market makers:

$$V_{ask}^{LOB} = \frac{1}{2}\left[\eta(1-\beta_{buy})(a-\sigma) + (1-\alpha_{buy})za\right],$$
$$V_{bid}^{LOB} = \frac{1}{2}\left[\eta(1-\beta_{sell})(b-\sigma) + (1-\alpha_{sell})zb\right],$$

[31] Following finance literature, we regard the arrivals of shocks as random events and abstract away from the possibility of correlated noise trading. Also, we keep the assumption that each noise trader is sufficiently small relative to the market size or total informed trading, which is a stylized fact established in the literature (see, for example, Easley and O'hara, 1987).

where the intuition is identical to the baseline model. The break-even prices are give by

$$a = a(\beta_{buy}, \alpha_{buy}) = \sigma \frac{(1-\beta_{buy})\eta}{(1-\beta_{buy})\eta + (1-\alpha_{buy})z}, \tag{31}$$

$$b = b(\beta_{sell}, \alpha_{sell}) = \sigma \frac{(1-\beta_{sell})\eta}{(1-\beta_{sell})\eta + (1-\alpha_{sell})z}. \tag{32}$$

Similarly, the indifference conditions for informed traders are modified as follows:

$$1 + a(\beta_{buy}, \alpha_{buy}) = \frac{p\left(\beta_{buy} + \frac{z}{2}\Delta\alpha\right)}{1-\tau}, \tag{33}$$

$$1 - b(\beta_{sell}, \alpha_{sell}) = \frac{p\left(-\beta_{sell} + \frac{z}{2}\Delta\alpha\right)}{1+\tau}, \tag{34}$$

where $\Delta\alpha \equiv \alpha_{buy} - \alpha_{sell}$. The marginal prices on the AMM depend on the difference between α_{buy} and α_{sell}, as noise traders always buy $\frac{z}{2}\Delta\alpha$ measure of the asset in aggregate.

5.2.2 Threshold Condition for Noise Traders

Noise traders compute the expected trading cost on the AMM by incorporating the possibility of informed trading. More precisely, a type-γ noise buyer expects to earn[32]

$$\pi^{AMM}_{N,buy}(\gamma) = \gamma \left[\frac{\eta}{2} \left(2 - \frac{P(\beta_{buy} + \Delta\alpha\frac{z}{2}) + P(-\beta_{sell} + \Delta\alpha\frac{z}{2})}{1-\tau} \right) + (1-\eta) \left(1 - \frac{P(\Delta\alpha\frac{z}{2})}{1-\tau} \right) \right].$$

The first term represents the case that the common-value shock occurs, and the noise trader trades together with informed traders. Conversely, the second term captures the case with no common-value shocks, where only noise traders trade. Similarly, a seller

[32]The expected profit involves the average price P rather than the marginal price p. This is because of the implicit assumption that all noise traders submit orders simultaneously, and they are processed following a uniform distribution. If a noise trader's order is executed after δ measures of trades have been completed, then her infinitesimal trade is executed at $p(\delta, X)$ in equation (9). Since $\delta \sim U[0, \bar{\delta}]$ with $\bar{\delta} \equiv \beta_{buy} + \frac{z}{2}\Delta\alpha$, the expected marginal execution price is $\frac{1}{\bar{\delta}} \int_0^{\bar{\delta}} p(\delta, X) d\delta = P(\bar{\delta}, X)$, as defined by equation (8).

expects to earn

$$\pi^{AMM}_{N,sell}(\gamma) = \gamma \left[\frac{\eta}{2} \left(\frac{P(\beta_{buy} + \Delta\alpha\frac{z}{2}) + P(-\beta_{sell} + \Delta\alpha\frac{z}{2})}{1+\tau} - 2 \right) + (1-\eta) \left(\frac{P(\Delta\alpha\frac{z}{2})}{1+\tau} - 1 \right) \right].$$

Comparing with the bid and ask prices on the LOB, noise traders' threshold conditions lead to the following equations:

$$\bar{\gamma}\alpha_{buy} \left[\frac{\eta}{2} \left(2 - \frac{P(\beta_{buy} + \Delta\alpha\frac{z}{2}) + P(-\beta_{sell} + \Delta\alpha\frac{z}{2})}{1-\tau} \right) + (1-\eta) \left(1 - \frac{P(\Delta\alpha\frac{z}{2})}{1-\tau} \right) \right] = -a, \tag{35}$$

$$\bar{\gamma}\alpha_{sell} \left[\frac{\eta}{2} \left(\frac{P(\beta_{buy} + \Delta\alpha\frac{z}{2}) + P(-\beta_{sell} + \Delta\alpha\frac{z}{2})}{1+\tau} - 2 \right) + (1-\eta) \left(\frac{P(\Delta\alpha\frac{z}{2})}{1+\tau} - 1 \right) \right] = -b, \tag{36}$$

where a and b are given by (31) and (32). Conditions (33)–(36) characterize traders' venue choice in the equilibrium and pin down $(\alpha_i, \beta_i)_{i=buy,sell}$.

5.2.3 Results and Intuition

The comparative statics of the solutions are not analytically obtained; instead, we conduct numerical experiments. The outcomes are detailed in Appendix D, illustrating both the positive liquidity spillover effect and the hump-shaped reaction of equilibrium AMM liquidity to asset volatility.

The core findings remain robust despite this extension, as our central argument survives: noise traders experience a smaller price impact on the AMM than informed traders. To grasp the intuition behind this, consider informed and noise traders buying the asset (i.e., equations [33] and [35]). Noise traders engage in trading in both cases, when the common-value shock occurs and when it does not. However, their trading cost depends on the nature of the trigger event: it is characterized by the trading volumes $(\beta_{buy} + \frac{z}{2}\Delta\alpha)$ and $\frac{z}{2}\Delta\alpha$ with probability η and $1-\eta$, respectively. Only in the case with the common-value shock, do noise traders trade with informed traders and incur a large price impact. Conversely, informed buyers always face the cost associated with volume $(\beta_{buy} + \frac{z}{2}\Delta\alpha)$,

leading to a higher expected price impact than noise traders. As a result, the heterogeneous reactions among these traders to variations in market parameters survive in this extension, supporting our main results.

5.3 The Latest AMM Designs: Uniswap v3 and v4

This section delves into the discussions surrounding the most recent and prospective versions of AMMs proposed and implemented by Uniswap.[33]

5.3.1 Concentrated Liquidity

Uniswap v3 has implemented an update on the baseline CPMM, known as the *concentrated liquidity* AMM.[34] The entire price range is discretized into ticks, denoted as $\mathcal{P} \equiv \{p_j\}_{j=-n}^{n}$, and each liquidity provider chooses a range of the price to supply liquidity, denoted as $s(l,u) \equiv [p_l, p_u]$ with $l < u$. A liquidity provider with strategy $s(l,u)$ is characterized by the liquidity parameter $k(l,u)$ in equation (B-32) in Appendix B.3. She earns fee income and incurs impermanent loss as long as the asset's price remains within the specified range $s(l,u)$. Once the price moves out of $s(l,u)$, her liquidity becomes inactive and no further returns accrue. Thus, liquidity provider i's contribution to a unit range $s(j,j+1)$ is defined by $k_{i,j}(l,u) = k(l,u)\mathbb{I}_{\{s(j,j+1)\subset s(l,u)\}}$, where $\mathbb{I}_D$ denotes the indicator function of event D. Also, her liquidity share in this range is $k_{i,j}(l,u)/K_j$ with aggregate liquidity $K_j \equiv \int k_{i,j}(l,u)di$. In subsequent discussions, we denote $\boldsymbol{K} = (K_j)_{j=-n}^{n}$ as the measure of AMM liquidity, which corresponds to X in the baseline model.

Given K_j, the liquidity pool in range $s(j,j+1)$ must follow the truncated (or shifted)

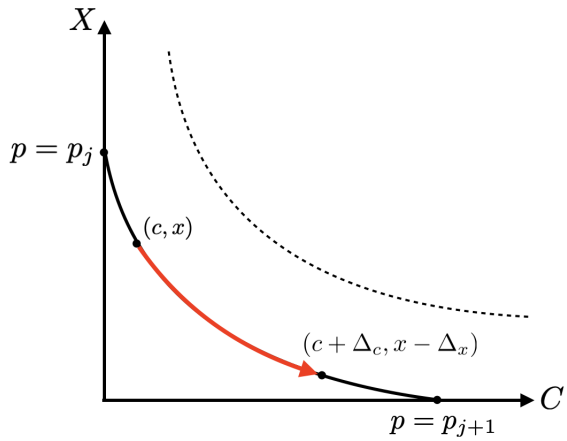
[33] See white papers by **?** for Uniswap v3 and Adams et al. (2023) for v4.

[34] Kyle and Lee (2017) propose a fully continuous market by introducing the concept of scale limit orders, which resembles the concentrated liquidity algorithm. It renders the demand and supply monotonic functions in the price, thereby eliminating a positive spread, as long as these functions intersect. The subsequent discussion demonstrates that liquidity providers offer liquidity to ensure the existence of such intersections in equilibrium due to the pro-rata reward allocation to liquidity providers.

bonding curve, as specified in the equation below and illustrated by Figure 7.

$$K_j^2 = \left(C + K_j \sqrt{p_j}\right)\left(X + \frac{K_j}{\sqrt{p_{j+1}}}\right). \tag{37}$$

Figure 7: Bonding Curve for Concentrated Liquidity



Note: The solid line illustrates the bonding curve in equation (37), while the dashed line represents the bonding curve in the baseline AMM. When an incoming swap order buys Δ_x of the asset, the liquidity pool moves downward along the curve, thereby increasing the marginal price (i.e., the inverse slope of the curve).

As in the baseline AMM, the marginal asset's price in $s(j, j+1)$ is determined by the inverse slope of the curve at the current liquidity pool, $(C, X) = (c, x)$, i.e., $p = -\frac{dC}{dX}|_{(c,x)}$. If the price moves out from $s(j, j+1)$ to the adjacent range, the same algorithm with $K_{j\pm1}$ determines the price .

Without loss of generality, we assume that $p_0 = 1$, $p_n = 1 + \sigma$, and $p_{-n} = 1 - \sigma$.[35] Also, the structures of the model other than the AMM are the same as the baseline model. The following analyses argue that two important mechanisms of the baseline model remain unchanged in this update: the marginal price satisfies Lemma 1, and the AMM does not involve a positive spread for a small trade. Note that the latter condition corresponds to strictly positive liquidity allocated to price ranges adjacent to p_0, i.e., $K_{\pm1} > 0$.

[35] As $\tilde{v} = 1 \pm \sigma$, and the equilibrium asset's price falls within interval $(1 - \sigma, 1 + \sigma)$, liquidity providers have no incentive to allocate liquidity outside of this range.

5.3.2 Liquidity Spillover Effect

Firstly, note that the liquidity spillover effect requires the marginal price $p(\delta, \boldsymbol{K})$ to satisfy conditions (i) and (ii) in Lemma 1: it increases with trading size ($\frac{\partial p(\delta,\boldsymbol{K})}{\partial \delta} > 0$), and the price impact decreases with the size of liquidity ($\frac{\partial p(\delta,\boldsymbol{K})}{\partial K_j} < 0$ if and only if $\delta > 0$). We first assume $K_j > 0$ for $l \leq j \leq u$ and verify this feature in the equilibrium.

Proposition 4. *Define price ticks p_u and p_l as*

$$(1 + a(\alpha, \beta_{buy}))(1 - \tau) = p(\beta_{buy}, K_u) \in s(u, u+1), \tag{38}$$

$$(1 - b(\alpha, \beta_{sell}))(1 + \tau) = p(-\beta_{sell}, K_l) \in s(l, l+1). \tag{39}$$

(i) If liquidity K_j increases for $l \leq j \leq u$, the bid-ask spread in the LOB declines.
(ii) Changes in K_j for $j < l$ and $j > u$ have no impact on the bid-ask spread.

Proof. See Appendix B.3. □

Since the marginal price in $s(j, j+1)$ follows the shifted bonding curve in (37), the original AMM's features persist in range $s(j, j+1)$ as long as $K_j > 0$. Therefore, the liquidity spillover effect arises in the relevant price range if $K_j > 0$ for $l \leq j \leq u$.

Secondly, following Aoyagi, Ip, Kawaguchi, Kuramoto and Tsuchida (2024), we argue that liquidity providers cover the relevant price ranges in equilibrium, i.e. $K_j > 0$ for $l \leq j \leq u$, thus eliminating a positive bid-ask spread and supporting the conclusion in Proposition 4.[36] Denoting the profits of liquidity provider i from allocating liquidity in range $s(l, u)$ as $V(l, u)$, Aoyagi et al. (2024) shows the following expression.

Lemma 9. *(i) The expected profit from taking $s(l, u)$ is expressed as the weighted average of profits from all single-unit ranges in this interval:*

$$V(l, u) = \sum_{j=l}^{u-1} \theta_j(l, u) V(j, j+1), \tag{40}$$

[36] In practice, the distribution of liquidity on Uniswap v3 typically centers around the current price (p_0) and covers a sufficiently wide range of price ticks, except for pairs of stablecoins. See data on liquidity distributions provided by Uniswap Analytics.

where $\theta_j(l,u) \equiv \frac{k(l,u)}{k(j,j+1)}$ *and* $\sum_j \theta_j(l,u) = 1$.
(ii) $V(j,j+1)$ *consists of a profit proportional to* $\frac{k_{i,j}(l,u)}{K_j}$.

Proof. See Appendix A of Aoyagi et al. (2024). □

While point (i) is an algorithmic feature of concentrated liquidity, point (ii) indicates strategic substitution among liquidity providers: as a reward for liquidity provision is allocated by the pro-rata rule, each liquidity provider tries to avoid a price range with a large K_j.

Suppose that a continuum of liquidity providers competitively supply liquidity by adjusting $s(l,u)$ to maximize their profits $V(l,u)$. Since they are price takers (i.e., they take $\boldsymbol{K}$ as given), Lemma 9 yields the following result.

Proposition 5. *In equilibrium, all relevant price ranges have strictly positive liquidity, i.e.,* $K_j > 0$ *for all* $l \leq j \leq u$.

Proof. See Appendix B.3. □

Intuitively, the weighted average expression in (40) and strategic substitution among liquidity providers imply that each liquidity provider must be indifferent between all unit price ranges in her liquidity allocation, i.e., $V(i,i+1) = V(j,j+1)\ \forall i,j \in [l,u]$. If the equilibrium allocation involves $K_m = 0$ for $m \in [l,u]$, re-allocating all liquidity to $s(m,m+1)$ becomes a strictly dominant strategy due to $\lim_{K_m=0} V(m,m+1) = \infty$, contradicting to $K_m = 0$. Therefore, Proposition 5 holds in the equilibrium, demonstrating that the concentrated liquidity AMM leads to a dispersed liquidity allocation and does not involve a positive spread for an infinitesimal trade.[37]

It is worth noting that the above result hinges on an additional distinctive feature of the AMM: the reward for liquidity provision is distributed on a pro-rata basis. As a result, competition among liquidity providers exhibits strategic substitution in terms of price ranges, leading to the above results. This market structure highlights a stark contrast to the LOB, in which a single market maker at the top of the book earns the

[37] The specific configuration of the liquidity distribution among price ranges is not determined from this analysis.

reward according to the price-time priority. LOB market makers undercut each other's quotes, and their competition involves strategic complementarity.

5.3.3 Contingent Fee

Uniswap v4 allows liquidity providers to charge contingent fee levels based on the characteristics of incoming traders.[38] This aims to charge higher fees to informed (toxic) traders and lower fees to noise (non-toxic) traders, thereby mitigating adverse selection and attracting more liquidity.

Our mechanism indicates that the baseline results are robust as long as noise traders are not required to pay a liquidity cost that *positively* depends on adverse selection. The new fee structure in Uniswap v4 tends to lower the liquidity cost (fees) charged to noise traders, as liquidity providers can earn higher compensation from informed trading. Therefore, the reaction of liquidity costs for noise traders to adverse selection is expected to be (weakly) negative, and the heterogeneous reactions of noise and informed traders should persist in the new environment.

5.4 Implication for Platform Design

The liquidity spillover effect in Proposition 1 has significant implications for the strategy and design of exchange platforms, including those for traditional securities. For instance, the NYSE could introduce AMM trading pools for stocks listed on its LOB platform to leverage the positive spillover effect and decrease LOB trading costs. The liquidity spillover effect in our model suggests a robust and predictable liquidity interaction, which might be more nuanced and challenging to predict in the case of parallel LOBs. However, this process may involve asymmetric changes between the bid and ask sides, resulting in a biased midpoint, as illustrated by Lemma 4.

Moreover, the hump-shaped impact of σ on X^* is relevant to the design of DEXs. For instance, the flexible fee structure in Uniswap v4 aims to encourage liquidity provision

[38] As of April 2024, the implementation of this update is scheduled for Q3 2024, so actual player activity in the updated version has not yet been observed. The following discussion is a conceptual argument based on the published source code for the update.

by allowing liquidity providers to charge higher fees when they anticipate higher asset volatility. However, there is no consensus on the optimal fee strategy, as evidenced by discussions on Uniswap's governance forum. Anecdotal evidence suggests that liquidity providers of volatile token pairs do not always choose pools with high fees.[39] This suggests that high volatility does not necessarily result in low market-making returns, as toxic (informed) investors and non-toxic retail (noise) traders respond differently to asset volatility, leading to a complex reaction of market maker profits. Our model indicates that a marginal increase in volatility can benefit AMM liquidity providers under certain parameter conditions, and the profit-maximizing fee level may depend on these parameters when considering traders' platform choice among different market-making structures.

6 Conclusion

This study investigates the equilibrium impact of integrating Automated Market Makers (AMMs), a novel market-making algorithm on decentralized exchanges (DEXs), into the market environment that is predominantly governed by limit-order books (LOBs). Through inducing disproportionate reactions from informed and noise traders, the new market structure generates a positive spillover effect on the liquidity of LOBs. Moreover, our findings offer new insights into the reaction of AMM liquidity to asset volatility and how the information content of order flow on LOBs is influenced by the presence of AMMs.

For the further development of our research, it is essential to analyze the welfare implications of AMMs and the optimal AMM design. While our model cannot directly address this question due to its zero-sum nature (wherein defining optimality is non-trivial), our findings emphasize the need to consider coexisting market structures in such analyses. Neglecting the coexistence and traders' endogenous platform selection could lead to unintended outcomes, as our model highlights the significance of liquidity and price interactions between heterogeneous market structures.

[39] See, for example, Reuptaken's posts on the Uniswap governance forum.

References

Abadi, Joseph and Markus Brunnermeier, "Blockchain Economics," *Federal Reserve Bank of Philadelphia Working Paper Series*, 2022, (22-15).

Adams, Hayden, Moody Salem, Noah Zinsmeister, Sara Reynolds, Austin Adams, Will Pote, Mark Toda, Alice Henshaw, Emily Williams, and Dan Robinson, "Uniswap v4 Core," Uniswap 2023.

_ **, Noah Zinsmeister, and Dan Robinson**, "Uniswap v2 Core," Uniswap 2020.

Admati, Anat R and Paul Pfleiderer, "A theory of intraday patterns: Volume and price variability," *The Review of Financial Studies*, 1988, *1* (1), 3–40.

Angeris, Guillermo, Alex Evans, and Tarun Chitra, "When does the tail wag the dog? Curvature and market making," *arXiv preprint arXiv:2012.08040*, 2020.

_ **and Tarun Chitra**, "Improved Price Oracles: Constant Function Market Makers," *arXiv preprint arXiv:2003.10001*, 2020.

_ **, Hsien-Tang Kao, Rei Chiang, Charlie Noyes, and Tarun Chitra**, "An analysis of Uniswap markets," *Cryptoeconomic Systems Journal*, 2019.

Aoyagi, Jun, "Liquidity Provision by Automated Market Makers," *Working paper, SSRN 3674178*, 2020.

_ **, Wang Hei Ip, Kohei Kawaguchi, Wataru Kuramoto, and Shinya Tsuchida**, "Toxicity-Competitiveness Trade-off in Concentrated Liquidity Provision," *Available at SSRN*, 2024.

Barbon, Andrea and Angelo Ranaldo, "On The Quality Of Cryptocurrency Markets: Centralized Versus Decentralized Exchanges," *Swiss Finance Institute Research Paper Series*, 2024.

Bolton, Patrick, Tano Santos, and Jose A Scheinkman, "Cream-skimming in financial markets," *The Journal of Finance*, 2016, *71* (2), 709–736.

Brolley, Michael and Marius Zoican, "Liquid speed: A micro-burst fee for low-latency exchanges," *Journal of Financial Markets*, 2023, *64*, 100785.

Budish, Eric, "The Economic Limits of Bitcoin and the Blockchain," 2018.

Cao, David, Brett Hemenway Falk, and Gerry Tsoukalas, "Automated Market Makers and the Value of Adaptive Fees," *Available at SSRN*, 2023.

Capponi, Agostino and Ruizhe Jia, "The Adoption of Blockchain-based Decentralized Exchanges," *arXiv preprint arXiv:2103.08842*, 2021.

_ **,** _ **, and Shihao Yu**, "The Information Content of Blockchain Fees," *Available at SSRN*, 2022.

Chen, Long, Lin William Cong, and Yizhou Xiao, "A Brief Introduction to Blockchain Economics," *Information for Efficient Decision Making: Big Data, Blockchain and Relevance*, 2020.

Chiu, Jonathan and Thorsten V Koeppl, "The economics of cryptocurrency: Bitcoin and beyond," *Canadian Journal of Economics/Revue canadienne d'économique*, 2022, *55* (4), 1762–1798.

Chowdhry, Bhagwan and Vikram Nanda, "Multimarket trading and market liquidity," *The Review of Financial Studies*, 1991, *4* (3), 483–511.

Cong, Lin William, Ye Li, and Neng Wang, "Tokenomics: Dynamic adoption and valuation," *The Review of Financial Studies*, 2021, *34* (3), 1105–1155.

Easley, David and Maureen O'hara, "Price, trade size, and information in securities markets," *Journal of Financial Economics*, 1987, *19* (1), 69–90.

Evans, Alex, "Liquidity Provider Returns in Geometric Mean Markets," *arXiv preprint arXiv:2006.08806*, 2020.

Foley, Sean, Peter O'Neill, and Tālis J Putniņš, "A better market design? Applying 'Automated Market Makers' to traditional financial markets," *Working Paper*, 2023.

Foucault, Thierry, Marco Pagano, and Ailsa Röell, *Market liquidity: theory, evidence, and policy*, Oxford University Press, USA, 2013.

Glosten, Lawrence R, "Is the electronic open limit order book inevitable?," *The Journal of Finance*, 1994, *49* (4), 1127–1161.

— **and Paul R Milgrom**, "Bid, ask and transaction prices in a specialist market with heterogeneously informed traders," *Journal of Financial Economics*, 1985, *14* (1), 71–100.

Han, Jianlei, Shiyang Huang, and Zhuo Zhong, "Trust in DeFi: An Empirical Study of the Decentralized Exchange," *Available at SSRN 3896461*, 2022.

Harvey, Campbell R, Ashwin Ramachandran, and Joey Santoro, *DeFi and the Future of Finance*, John Wiley & Sons, 2021.

Hasbrouck, Joel, Thomas J Rivera, and Fahad Saleh, "The Need for Fees at a DEX: How Increases in Fees Can Increase DEX Trading Volume," *Available at SSRN 4192925*, 2023.

Huberman, Gur, Jacob D Leshno, and Ciamac Moallemi, "Monopoly without a monopolist: An economic analysis of the bitcoin payment system," *The Review of Economic Studies*, 2021, *88* (6), 3011–3040.

Kyle, Albert S, "Continuous auctions and insider trading," *Econometrica*, 1985, pp. 1315–1335.

— **and Jeongmin Lee**, "Toward a fully continuous exchange," *Oxford Review of Economic Policy*, 2017, *33* (4), 650–675.

Lehar, Alfred and Christine A Parlour, "Decentralized Exchanges," *The Journal of Finance*, 2024.

Malinova, Katya and Andreas Park, "Market Design with Blockchain Technology," *Available at SSRN 2785626*, 2017.

_ and _ , "Learning from DeFi: Would Automated Market Makers Improve Equity Trading?," *Available at SSRN 4531670*, 2024.

Menkveld, Albert J and Marius A Zoican, "Need for speed? Exchange latency and liquidity," *The Review of Financial Studies*, 2017, *30* (4), 1188–1228.

Milionis, Jason, Ciamac C Moallemi, Tim Roughgarden, and Anthony Lee Zhang, "Automated Market Making and Loss-versus-Rebalancing," *arXiv preprint arXiv:2208.06046*, 2024.

Pagano, Marco, "Trading volume and asset liquidity," *The Quarterly Journal of Economics*, 1989, *104* (2), 255–274.

Pagnotta, Emiliano and Andrea Buraschi, "An equilibrium valuation of bitcoin and decentralized network assets," *Available at SSRN 3142022*, 2018.

Park, Andreas, "The Conceptual Flaws of Constant Product Automated Market Making," *Management Science*, September 2023, *Special Section of Management Science: Blockchains and Crypto Economics*, 6731–6751.

Parlour, Christine A and Duane J Seppi, "Liquidity-based competition for order flow," *The Review of Financial Studies*, 2003, *16* (2), 301–343.

Schilling, Linda and Harald Uhlig, "Some simple bitcoin economics," *Journal of Monetary Economics*, 2019, *106*, 16–26.

Seppi, Duane J, "Liquidity provision with limit orders and a strategic specialist," *The Review of Financial Studies*, 1997, *10* (1), 103–150.

Warren, Will and Amir Bandeali, "0x: An open protocol for decentralized exchange on the Ethereum blockchain," *URl: https://github. com/0xProject/whitepaper*, 2017.

Xu, Jiahua, Krzysztof Paruch, Simon Cousaert, and Yebo Feng, "Sok: Decentralized exchanges (dex) with automated market maker (AMM) protocols," *arXiv preprint arXiv:2103.12732*, 2021.

Zhang, Yi, Xiaohong Chen, and Daejun Park, "Formal specification of constant product (xy=k) market maker model and implementation," *manuscript*, 2018.

Zhu, Haoxiang, "Do dark pools harm price discovery?," *The Review of Financial Studies*, 2014, *27* (3), 747–789.

Internet Appendix

A Notations

Table 1 summarizes notations in the baseline model.

Table 1: Summary of Notations

Notation	Definition
η	frequency of common-value shock
σ	size of common-value shock
α	fraction of noise traders on DEX
β_i	fraction of informed traders on DEX
z	trading size of a noise trader
τ	trade-execution fee on DEX
S	bid-ask spread on CEX
A_i	size of arbitrage trading in $t = 2$
X_i	$X_i = A_i + \beta_i$
X	total amount of asset supplied on DEX
C	total amount of cash supplied on DEX according to f
f	constant function for AMMs
$c = h(x)$	bonding curve on DEX
p_0	initial marginal price of the asset on DEX
p	marginal price on DEX for a small trade
P	cumulative price on DEX for liquidity providers
π_i^k	profits of agent i on platform k
ε	opportunity/participation cost for DEX liquidity providers

B Robustness Checks

In this section, we provide the further robustness checks and the proofs for the results in Section 5.

B.1 When Informed Trader Trades Multiple Units

In what follows, we provide a full characterization of the model in Subsection 5.1 and its proofs. Note that $\tilde{v} = 1 - \sigma$ leads the informed trader to sell the asset, where the total

benefit from selling $x_{L,sell}$ units in the LOB is given by

$$B(x_{L,sell}, \alpha) \equiv x_{L,sell} \left(1 - \sigma s(x_{L,sell}, \alpha)\right).$$

The marginal benefit is represented by

$$bid(x_{L,sell}) \equiv \frac{dB(x_{L,sell})}{dx_{L,sell}} = 1 - \sigma s(x_{L,sell}, \alpha)(2 - s(x_{L,sell}, \alpha))$$

and is decreasing in the trading volume: $\frac{1}{\sigma}\frac{\partial bid(x_{L,buy})}{\partial x_{L,sell}} = -2\sigma(1 - s(x_{L,sell}, \alpha))\frac{\partial s(x_{L,sell},\alpha)}{\partial x_{L,sell}} < 0$.

The profit from selling the asset is

$$V_j = \frac{p(-x_{A,sell}, X)}{1+\tau} x_{A,sell} + B(x_{L,sell}) - (1-\sigma)x_{sell} - \frac{c}{2}x_{sell}^2,$$

leading to the following FOCs.

$$(x_{sell}) : -(1-\sigma) - cx_{sell} + bid(x_{L,sell}, \alpha) = 0, \tag{B-1}$$

$$(x_{A,sell}) : -\tilde{p}(-x_{A,sell}) + bid(x_{L,sell}, \alpha) = 0.$$

The condition (29) is derived by evaluating the first FOCs in (28) and (B-1) at $x_{L,j} = z$ and $\alpha = 0$, so that $x_{L,j} < z$ holds for all $\alpha > 0$.

Proof of Lemmas 7 and 8. Lemma 8 directly follows from Lemma 2 by rewriting $\beta_j = x_{A,j}$ and normalizing $x_j = 1$.

As for Lemma 7, consider the following condition.

$$(1-\eta)^2 < 1 - \frac{\tau}{\sigma(1-\tau)}. \tag{B-2}$$

As the analyses below attest, if this condition is violated, $x_{A,j} = 0$ becomes an equilibrium, leading to a single-platform environment. Otherwise, this condition ensures that $x_{A,j} > 0$ by preventing the relative marginal price on the AMM from being extremely high. The following analyses assume that inequality (B-2) holds to focus on non-trivial cases.

For $j = buy$ and $sell$, the first FOCs in (28) and (B-1) lead to

$$x_j = \frac{\sigma}{c}\left(1 - s(x_{L,j}, \alpha)\right)^2, \tag{B-3}$$

and the FOCs are summarized by $x_{L,j}$ and α:

$$\begin{cases} \frac{p\left(\frac{\sigma}{c}\left(1-s(x_{L,buy},\alpha)\right)^2 - x_{L,buy}, X\right)}{1-\tau} = 1 + \sigma - \sigma\left(1 - s(x_{L,buy}, \alpha)\right)^2 & \text{for } j = buy, \\ \frac{p\left(-\frac{\sigma}{c}\left(1-s(x_{L,sell},\alpha)\right)^2 + x_{L,sell}, X\right)}{1+\tau} = 1 - \sigma + \sigma\left(1 - s(x_{L,sell}, \alpha)\right)^2 & \text{for } j = sell. \end{cases} \tag{B-4}$$

In the first condition, the LHS is decreasing with $x_{L,buy}$ and α, while the RHS is increasing with these variables. Since LHS $>$ RHS at $x_{L,buy} = 0$ and LHS $<$ RHS at $x_{L,buy} \to z$ if (B-2) holds, the first equation of (B-4) has a unique solution $x_{L,buy} = x_{L,buy}(\alpha; X)$, which is decreasing in α due to the implicit function theorem. Analogously, there is a unique solution, $x_{L,sell}$, to the second FOC, which is decreasing in α due to the implicit function theorem.

To understand the reaction of x_j and $x_{A,j}$, take (B-4) as an equation of s and $x_{L,j}$. Since $x_{L,buy}$ decreases with α, $s(x_{L,buy}, \alpha)$ is increasing in α incorporating the reaction of $x_{L,buy}$. Therefore, from (B-3) and the first condition of (B-4), x_{buy} decreases and $x_{A,buy}$ increases with α. The symmetric argument holds for the sell orders.

Finally, we observe that $x_{L,buy}$ is the solution to the following equation with respect to x:

$$\frac{p\left(\frac{\sigma}{c}\left(1 - s(x)\right)^2 - x\right)}{1-\tau} - 1 = \sigma\left[1 - \left(1 - s(x)\right)^2\right], \tag{B-5}$$

while $x_{L,sell}$ is the solution to

$$1 - \frac{p\left(-\frac{\sigma}{c}\left(1 - s(x)\right)^2 + x\right)}{1+\tau} = \sigma\left[1 - \left(1 - s(x)\right)^2\right]. \tag{B-6}$$

Due to the convexity of p, it holds that

$$p\left(\frac{\sigma}{c}\left(1 - s(x)\right)^2 - x\right) - 1 > 1 - p\left(-\frac{\sigma}{c}\left(1 - s(x)\right)^2 + x\right).$$

Therefore, incorporating $\frac{1}{1-\tau} > \frac{1}{1+\tau}$, it must hold from (B-5) and (B-6) that

$$1-(1-s(x_{L,buy}))^2 > 1-(1-s(x_{L,sell}))^2,$$

implying that $x_{L,buy} > x_{L,sell}$. This inequality and equation (B-3) indicate that $x_{buy} < x_{sell}$. Since $x_{A,j} = x_j - x_{L,j}$, we conclude that $x_{A,sell} > x_{A,buy}$. □

Proof of Proposition 3. By incorporating the reactions of $x = x_{L,j}$ to X, the partial derivative of s in (B-7) with respect to X is

$$\frac{\partial s(x,\alpha)}{\partial X} = \eta(1-\eta)z\frac{(1-\alpha)\frac{\partial x}{\partial X} + x\frac{\partial \alpha}{\partial X}}{T^2(x)}, \tag{B-7}$$

where $T(x) \equiv \eta x + z(1-\eta)(1-\alpha)$. We denote $T_j = T(x_{L,j})$ hereafter. Also, (B-3) indicates that

$$\frac{\partial x_j}{\partial X} = -2\frac{\sigma}{c}\left(1-s(x_{L,j},\alpha)\right)\frac{\partial s(x_{L,j},\alpha)}{\partial X}. \tag{B-8}$$

Furthermore, the marginal price on the AMM exhibits the following reaction to X; for a buy order,

$$\begin{aligned}\frac{dp(x_{A,buy},X)}{dX} &= \frac{\partial p(x_{A,buy},X)}{\partial X} + \left(\frac{\partial x_{buy}}{\partial X} - \frac{\partial x_{L,buy}}{\partial X}\right)\frac{\partial p(x_{A,buy},X)}{\partial x_{A,buy}} \\ &= \frac{\partial p(x_{A,buy},X)}{\partial X} - \left[R_{buy}x_{L,buy}\frac{\partial \alpha}{\partial X} + \left(1+R_{buy}(1-\alpha)\right)\frac{\partial x_{L,buy}}{\partial X}\right]\frac{\partial p(x_{A,buy},X)}{\partial x_{A,buy}},\end{aligned} \tag{B-9}$$

and for a sell order;

$$\begin{aligned}\frac{dp(-x_{A,sell},X)}{dX} &= \frac{\partial p(-x_{A,sell},X)}{\partial X} - \left(\frac{\partial x_{sell}}{\partial X} - \frac{\partial x_{L,sell}}{\partial X}\right)\frac{\partial p(-x_{A,sell},X)}{\partial x_{A,sell}} \\ &= \frac{\partial p(-x_{A,sell},X)}{\partial X} + \left[R_{sell}x_{L,sell}\frac{\partial \alpha}{\partial X} + \left(1+R_{sell}(1-\alpha)\right)\frac{\partial x_{L,sell}}{\partial X}\right]\frac{\partial p(-x_{A,sell},X)}{\partial x_{A,sell}},\end{aligned} \tag{B-10}$$

where

$$R_j \equiv \frac{2\sigma\eta(1-\eta)z}{cT_j^2}(1 - s(x_{L,j}, \alpha)).$$

Applying the partial derivative to the threshold condition for noise traders in (30), we obtain

$$\left(\frac{2\bar{\tau}\bar{\gamma}}{\sigma\eta(1-\eta)z} - \sum_{j=buy,sell} \frac{x_{L,j}}{T_j^2}\right)\frac{\partial\alpha}{\partial X} = (1-\alpha)\sum_{j=buy,sell}\frac{1}{T_j^2}\frac{\partial x_{L,j}}{\partial X}. \tag{B-11}$$

Also, by taking the partial derivative of equations in (B-4), the following conditions hold.

$$\frac{1}{1-\tau}\frac{dp(x_{A,buy}, X)}{dX} = 2\sigma\left(1 - s(x_{L,buy}, \alpha)\right)\frac{\partial s(x_{L,buy}, \alpha)}{\partial X}, \tag{B-12}$$

$$\frac{1}{1+\tau}\frac{dp(-x_{A,sell}, X)}{dX} = -2\sigma\left(1 - s(x_{L,sell}, \alpha)\right)\frac{\partial s(x_{L,sell}, \alpha)}{\partial X}. \tag{B-13}$$

Adopting equations (B-7)-(B-10) and rearranging the terms, we obtain

$$\frac{\partial x_{L,buy}}{\partial X} = \frac{\frac{1}{1-\tau}\frac{\partial p(x_{A,buy},X)}{\partial X} - \left(\frac{1}{1-\tau}\frac{\partial p(x_{A,buy},X)}{\partial x_{A,buy}} + c\right)R_{buy}x_{L,buy}\frac{\partial\alpha}{\partial X}}{c(1-\alpha)R_{buy} + \frac{1}{1-\tau}\left(1 + (1-\alpha)R_{buy}\right)\frac{\partial p(x_{A,buy},X)}{\partial x_{A,buy}}}, \tag{B-14}$$

$$\frac{\partial x_{L,sell}}{\partial X} = -\frac{\frac{1}{1+\tau}\frac{\partial p(-x_{A,sell},X)}{\partial X} + \left(\frac{1}{1+\tau}\frac{\partial p(-x_{A,sell},X)}{\partial x_{A,sell}} + c\right)R_{sell}x_{L,sell}\frac{\partial\alpha}{\partial X}}{c(1-\alpha)R_{sell} + \frac{1}{1+\tau}\left(1 + (1-\alpha)R_{sell}\right)\frac{\partial p(-x_{A,sell},X)}{\partial x_{A,sell}}}. \tag{B-15}$$

By plugging the above expressions into (B-11),

$$J\frac{\partial\alpha}{\partial X} = \frac{1}{T_{buy}^2}\frac{\frac{1}{1-\tau}\frac{\partial p(x_{A,buy},X)}{\partial X}}{\left(c + \frac{1}{1-\tau}\frac{\partial p(x_{A,buy},X)}{\partial x_{A,buy}}\right)(1-\alpha)R_{buy} + \frac{1}{1-\tau}\frac{\partial p(x_{A,buy},X)}{\partial x_{A,buy}}} \tag{B-16}$$

$$- \frac{1}{T_{sell}^2}\frac{\frac{1}{1+\tau}\frac{\partial p(-x_{A,sell},X)}{\partial X}}{\left(c + \frac{1}{1+\tau}\frac{\partial p(-x_{A,sell},X)}{\partial x_{A,sell}}\right)(1-\alpha)R_{sell} + \frac{1}{1+\tau}\frac{\partial p(-x_{A,sell},X)}{\partial x_{A,sell}}}, \tag{B-17}$$

where

$$J \equiv \frac{1}{(1-\alpha)}\left(\frac{2\bar{\tau}\bar{\gamma}}{\sigma\eta(1-\eta)z} - \sum_{j=buy,sell}\Delta_j\right),$$

with

$$\Delta_{buy} = \frac{x_{L,buy}}{T_{buy}^2} \frac{\frac{1}{1-\tau}\frac{\partial p(x_{A,buy},X)}{\partial x_{A,buy}}}{\left(c + \frac{1}{1-\tau}\frac{\partial p(x_{A,buy},X)}{\partial x_{A,buy}}\right)(1-\alpha)R_{buy} + \frac{1}{1-\tau}\frac{\partial p(x_{A,buy},X)}{\partial x_{A,buy}}},$$

and

$$\Delta_{sell} = \frac{x_{L,sell}}{T_{sell}^2} \frac{\frac{1}{1+\tau}\frac{\partial p(-x_{A,sell},X)}{\partial x_{A,sell}}}{\left(c + \frac{1}{1+\tau}\frac{\partial p(-x_{A,sell},X)}{\partial x_{A,sell}}\right)(1-\alpha)R_{sell} + \frac{1}{1+\tau}\frac{\partial p(-x_{A,sell},X)}{\partial x_{A,sell}}}.$$

Since condition (ii) in Lemma 1 suggests that

$$\frac{\partial p(x_{A,buy}, X)}{\partial X} < 0 \text{ and } \frac{\partial p(-x_{A,sell}, X)}{\partial X} > 0,$$

the RHS of (B-16) is negative. As for the LHS, note that

$$\begin{aligned}(1-\alpha)J &> \frac{2\bar{\tau}\bar{\gamma}}{\sigma\eta(1-\eta)z} - \sum_{j=buy,sell} \frac{x_{L,j}}{T_j^2} \\ &> 2\left(\frac{\bar{\tau}\bar{\gamma}}{\sigma} - \frac{1}{1-\alpha}\right)\frac{1}{\eta(1-\eta)}.\end{aligned}$$

Following the same logic as the baseline model (see Appendix C.3), $J > 0$ holds. Therefore, we conclude that α is decreasing in X, which in turn indicates that the bid-ask spread shrinks as X increases. □

The comparative statics regarding asset volatility are provided in Figures 8 and 9, where we confirm the hump-shaped reaction of the AMM liquidity to σ.

B.2 Alternative Fee Structure

According to the algorithm in Uniswap v2, we consider the alternative assumption regarding the transaction fee (see Angeris and Chitra, 2020 for more details):

Assumption 1. *Assume that the AMM deals with the fee payment with the following rules:*
(i) The fee is paid in term of a means of payment;
(ii) The pool balance is updated by incorporating the fee; and
(iii) The execution price is computed based on the post-trade pool without adding the fee.

It is shown in the literature (e.g., Angeris and Chitra, 2020) that Assumption 1 makes the price *path dependent*, i.e., the total payment differs when a trader executes a δ-units of buy order all at once versus splitting her order.

In what follows, we focus on the CPMM to derive the analytical solution and show that the marginal price function, p, under Assumption 1 satisfies all conditions in Lemma 1. While the path dependence may affect our quantitative results, all qualitative results in the baseline model are robust to this change.

B.2.1 Formalizing Assumption 1

To formalize Assumption 1, consider the initial liquidity pool with (X, C), where $C = X$ due to the non-arbitrage condition, and the initial constant $k = X^2$. According to rule (i), if an incoming order buys Δ_x units of the asset by paying Δ_c of cash, it must pay $\tau\Delta_c$ of cash as a fee, while an order buying Δ_c of cash by adding Δ_x of the asset must pay $\tau\Delta_x$ as a fee. Moreover, rules (ii) and (iii) imply that the pool balance after the trade becomes $(X - \Delta_x, X + \Delta_c)$, while the execution price for this order is computed using the bonding curve

$$k = X^2 = (X - \Delta_x)(X + (1 - \tau)\Delta_c).$$

Subsequently, the execution price for the next incoming trade is computed using the updated constant, $k' = (X - \Delta_x)(X + \Delta_c)$.

B.2.2 Marginal Price for Informed Traders

Consider the marginal price of the asset in the AMM for an incoming informed trader. Prior to this trader's arrival, assume that other informed traders have completed their trading with a total volume δ. To analyze the impact of Assumption 1, we first discretize this volume by considering n informed traders, each trading $x \equiv \delta/n$ and indexed by $i = 1, 2, \cdots, n$. Note that our baseline model with a continuum of traders can be approximated by taking the limit as $n \to \infty$ with δ being constant. In what follows, we allow $x < 0$ and $\delta < 0$ to represent traders selling the asset.

Lemma 10. *Suppose that informed traders have collectively traded δ units of the asset, and denote*

the marginal price for the next incoming informed trader as $p(\delta, X)$. This marginal price satisfies all conditions stipulated in 1 of the baseline model.

Proof. Consider the case with $\tilde{v} = 1 + \sigma$ and informed traders buying the asset. Denote the cash payment of the i-th trader as $g_i(x, X)$.

According to Assumption 1, the first trader must pay g_1 to achieve the following bonding curve:

$$(X - x)(X + (1 - \tau)g_1(x, X)) = X^2.$$

$$\therefore g_1(x, X) = \frac{1}{1 - \tau}\frac{xX}{X - x}.$$

Similarly, the second trader's payment g_2 is determined by

$$(X - 2x)(X + g_1(x, X) + g_2(x, X)) = (X - x)(X + g_1(x, X)).$$

$$\therefore g_2(x, X) = \frac{1}{1 - \tau}\frac{X(X + g_1(x, X))}{X - 2x}.$$

Note that the pool balance is updated by incorporating the fee payment due to rules (ii) and (iii). By induction, we obtain

$$g_i(x, X) = \frac{1}{1 - \tau}\frac{x(X + \sum_{j=1}^{i-1} g_j(x, X))}{X - ix}. \tag{B-18}$$

Based on (B-18), compute the marginal price of the asset when the i-th trader arrives at the market:

$$p_i(x, X) \equiv \lim_{\Delta \to 0} \frac{\partial}{\partial \Delta}\left(\frac{1}{1 - \tau}\frac{\Delta(X + \sum_{j=1}^{i-1} g_j(x, X))}{X - (i - 1)x - \Delta}\right) \tag{B-19}$$

$$= \frac{1}{1 - \tau}\frac{X + \sum_{j=1}^{i-1} g_j(x, X)}{X - (i - 1)x}, \tag{B-20}$$

where the inside of the parentheses in equation (B-19) represents the total payment of the i-th trader if she trades Δ units of the asset. Equations (B-18) and (B-20) lead to the following expression:

$$g_i(x, X) = \frac{X - (i - 1)x}{X - ix} x p_i(x, X). \tag{B-21}$$

Applying the same logic to the marginal price for trader $i+1$, we obtain the following recursive expression:

$$\begin{aligned} p_{i+1}(x,X) &= \frac{1}{1-\tau}\frac{X+\sum_{j=1}^{i-1} g_j(x,X)+g_i(x,X)}{X-ix} \\ &= \underbrace{\frac{X-(i-1)x}{X-ix}\left(1+\frac{1}{1-\tau}\frac{x}{X-ix}\right)}_{\equiv r_i(x,X)} p_i(x,X). \end{aligned} \tag{B-22}$$

Note that we are interested in $p(\delta, X) \equiv p_{n+1}(x, X)$.

In equation (B-22), coefficient $r_i(x, X)$ has the following properties: $\frac{\partial r_i}{\partial x} > 0$, $\frac{\partial^2 r_i}{\partial x^2} > 0$, and $\frac{\partial r_i}{\partial X} < 0$ for $x > 0$. Furthermore, taking the partial derivative of equation (B-22), the following equations hold.

$$\frac{\partial p_{i+1}(x,X)}{\partial x} = r_i(x,X)\frac{\partial p_i(x,X)}{\partial x} + \frac{\partial r_i(x,X)}{\partial x} p_i(x,X), \tag{B-23}$$

$$\frac{\partial^2 p_{i+1}(x,X)}{\partial x^2} = r_i(x,X)\frac{\partial^2 p_i(x,X)}{\partial x^2} + 2\frac{\partial r_i(x,X)}{\partial x}\frac{\partial p_i(x,X)}{\partial x} + p_i(x,X)\frac{\partial^2 r_i(x,X)}{\partial x^2}, \tag{B-24}$$

and

$$\frac{\partial p_{i+1}(x,X)}{\partial X} = r_i(x,X)\frac{\partial p_i(x,X)}{\partial X} + \frac{\partial r_i(x,X)}{\partial X} p_i(x,X). \tag{B-25}$$

Since $p_1(x, X) = \frac{1}{1-\tau}$, we have $\frac{\partial p_1(x,X)}{\partial x} = \frac{\partial p_1(x,X)}{\partial X} = 0$. Also, it holds that $\frac{\partial^2 p_2(x,X)}{\partial x^2} > 0$.[40] Incorporating the derivatives of $r_i(x, X)$, equations (B-23)-(B-25) imply that

$$\frac{\partial p(\delta,X)}{\partial \delta} > 0, \ \frac{\partial^2 p(\delta,X)}{\partial \delta^2} > 0, \text{ and } \frac{\partial p(\delta,X)}{\partial X} < 0.$$

Therefore, we conclude that all conditions in Lemma 1 hold for $x > 0$.

Next, consider the case that informed traders sell the asset due to $\tilde{v} = 1 - \sigma$. Suppose that each trader buys $x < 0$ units of the asset, meaning that she sells $-x > 0$ units. Note that traders pay their fees in terms of the asset due to rule (i). The first trader pays $g_1(x, X)$

[40] We consider the second-order derivative of p_2 because p_1 is not continuously differentiable.

so that

$$(X-(1-\tau)x)(X+g_1(x,X)) = X^2,$$

$$\therefore g_1(x,X) = \frac{(1-\tau)Xx}{X-(1-\tau)x}.$$

The second trader pays g_2 by following the updated bonding curve.

$$(X-x-(1-\tau)x)(X+g_1(x,X)+g_2(-x,X)) = (X-(1-\tau)x)(X+g_1(-x,X)),$$

$$\therefore g_2(x,X) = \frac{(1-\tau)x(X+g_1(x,X))}{X-x-(1-\tau)x}.$$

In general, the i-th trader's payment is given by

$$g_i(x,X) = \frac{(1-\tau)x(X+\sum_{j=1}^{i-1} g_j(x,X))}{X-(i-\tau)x}. \tag{B-26}$$

Based on the same arguments as the case of buying the asset, the marginal price for trader i when she starts executing her trade is expressed as

$$\begin{aligned} p_i(x,X) &= \lim_{\Delta\to 0}\frac{\partial}{\partial\Delta}\left(\frac{(1-\tau)\Delta(X+\sum_{j=1}^{i-1} g_j(x,X))}{X-(i-1)x-(1-\tau)\Delta}\right) \\ &= \frac{(1-\tau)(X+\sum_{j=1}^{i-1} g_j(x,X))}{X-(i-1)x}. \end{aligned}$$

Hence, equation (B-26) can be re-written as

$$g_i(x,X) = \frac{X-(i-1)x}{X-(i-\tau)x}xp_i.$$

Considering the marginal price for trader $i+1$, we obtain the recursive equation:

$$\begin{aligned} p_{i+1}(x,X) &= \frac{(1-\tau)(X+\sum_{j=1}^{i-1} g_j(x,X)) + (1-\tau)g_i(x,X)}{X-ix} \\ &= \underbrace{\frac{(X-(i-1)x)^2}{(X-ix)(X-(i-\tau)x)}}_{\equiv q_i(x,X)} p_i(x,X). \end{aligned} \tag{B-27}$$

Note that coefficient q satisfies $\frac{\partial q_i}{\partial x} > 0$, $\frac{\partial^2 q_i}{\partial x^2} > 0$, and $\frac{\partial q_i}{\partial X} > 0$ for $x < 0$. Since $\frac{\partial p_1(x,X)}{\partial x} = \frac{\partial p_1(x,X)}{\partial X} > 0$ and $\frac{\partial p_2}{\partial x^2} > 0$, equation (B-27) leads to the following inequalities:

$$\frac{\partial p(\beta, X)}{\partial \beta} > 0, \ \frac{\partial p^2(\beta, X)}{\partial \beta^2} > 0, \text{ and } \frac{\partial p(\beta, X)}{\partial X} > 0.$$

Hence, conditions in Lemma 1 hold in the case of selling the asset.

Finally, to explore the case with a continuum of informed traders, rewrite the recursive marginal price for buyers as

$$p_{n+1}(\frac{\delta}{n}, X) = \frac{X - \frac{\delta}{n}}{X - \frac{N\delta}{n}} \prod_{i=1}^{n} \left(1 + \frac{1}{1-\tau} \frac{\frac{\delta}{n}}{X - i\frac{\delta}{n}}\right) p_1(\frac{\delta}{n}, X).$$

At the limit of $n \to \infty$,

$$p(\delta, X) = \lim_{n\to\infty} (1-\tau) p_{n+1}(\frac{\delta}{n}, X) = \lim_{n\to\infty} \frac{X}{X-\delta} \prod_{i=1}^{n} \left(1 + \frac{1}{1-\tau} \frac{\frac{\delta}{n}}{X - i\frac{\delta}{n}}\right).$$

By applying the natural log to both sides,

$$\begin{aligned} \lim_{n\to\infty} \sum_{i=1}^{n} \log\left(1 + \frac{1}{1-\tau} \frac{\frac{\delta}{n}}{X - i\frac{\delta}{n}}\right) &= \lim_{n\to\infty} n \int_0^1 \log\left(1 + \frac{1}{1-\tau} \frac{\delta}{n} \frac{1}{X - t\delta}\right) dt \\ &= \frac{1}{(1-\tau)} \log\left(\frac{X}{X-\beta}\right). \end{aligned}$$

Thus, we obtain

$$p(\delta, X) = \left(\frac{X}{X-\delta}\right)^{\frac{2-\tau}{1-\tau}}. \tag{B-28}$$

For sellers, the analogous computation leads to

$$p(-\delta, X) = \left(\frac{X}{X+\delta}\right)^{2-\tau}. \tag{B-29}$$

Note that both marginal prices clearly satisfy the conditions in Lemma 1. □

B.2.3 Noise Traders

Consider sequential arrivals of noise traders with fixed α. We assume that buy and sell orders arrive alternatively, and the orders are executed according to the fee structure in Assumption 1.

Suppose that a noise trader arrives at the AMM after $\tilde{\alpha}$ measure of other traders have already completed i rounds of buy and sell orders, meaning that each order is of size $\frac{\tilde{\alpha}}{i}$. Following the analyses in the previous subsection, this trader faces the liquidity pool (C_i, X_i) with $X_i = X$ and

$$C_i = \left(1 + \frac{\tilde{\tau}}{X - \tau\frac{\tilde{\alpha}}{i}}\frac{\tilde{\alpha}}{i}\right)^i X$$

with $\tilde{\tau} = \frac{1-(1-\tau)^2}{1-\tau}$.

The marginal price for the incoming noise trader is computed based on (C_i, X_i). Hence, her expected marginal price, considering a buy and sell shocks happening with equal probability, is given by

$$\bar{p}_i = \frac{\tilde{\tau}}{2}\left(1 + \frac{\tilde{\tau}}{X - \tau\frac{\tilde{\alpha}}{i}}\frac{\tilde{\alpha}}{i}\right)^i.$$

We are interested in the expected marginal price when each noise trader is infinitesimal. By taking natural log and limit $i \to \infty$,

$$\begin{aligned}
\lim_{i\to\infty} \log \bar{p}_i &= \log\frac{\tilde{\tau}}{2} + \lim_{i\to\infty} i \log\left(1 + \frac{\tilde{\tau}\frac{\tilde{\alpha}}{i}}{X - \tau\frac{\tilde{\alpha}}{i}}\right) \\
&= \log\frac{\tilde{\tau}}{2} + \lim_{i\to\infty} i\left[\frac{\tilde{\tau}\frac{\tilde{\alpha}}{i}}{X - \tau\frac{\tilde{\alpha}}{i}} - \frac{i}{2}\left(\frac{\tilde{\tau}\frac{\tilde{\alpha}}{i}}{X - \tau\frac{\tilde{\alpha}}{i}}\right)^2 + \frac{i}{3}\left(\frac{\tilde{\tau}\frac{\tilde{\alpha}}{i}}{X - \tau\frac{\tilde{\alpha}}{i}}\right)^3 + \cdots\right] \\
&= \log\frac{\tilde{\tau}}{2} + \tilde{\tau}\frac{\tilde{\alpha}}{X}.
\end{aligned}$$

Hence, it holds that

$$\bar{p}_i = \bar{p}(\tilde{\alpha}) \equiv \frac{\tilde{\tau}}{2}e^{\tilde{\tau}\frac{\tilde{\alpha}}{X}}. \tag{B-30}$$

To incorporate the arrival timing of this trader, we assume that the noise traders' arrival process follows a uniform distribution. Consequently, the following result holds.

Lemma 11. *The ex-ante expected marginal trading cost for an AMM noise trader is expressed by the following:*

$$\begin{aligned}\frac{1}{\alpha}\int_0^{\alpha}\bar{p}(\tilde{\alpha})d\tilde{\alpha} &= \frac{1}{2}\frac{X}{\alpha}\left(e^{\tilde{\tau}\frac{\alpha}{X}}-1\right)\\ &\approx \frac{\tilde{\tau}}{2}.\end{aligned} \tag{B-31}$$

Proof. The first line directly follows from equation (B-30) and the uniform arrival. The last expression follows the fact that τ and $\tilde{\tau}$ are sufficiently small and the approximation $e^y \sim 1+y$ with sufficiently small y. □

Note that this expected price is identical to the baseline model by replacing $\bar{\tau}$ with $\tilde{\tau}$. In conclusion, the (expected) marginal trading costs for informed and noise traders satisfy all conditions in Lemma 1, and our main results are robust to the fee structure in Assumption 1.

B.2.4 When Informed Trader Trades Multiple Units

Related to the robustness check in Subsection 5.1, the following lemma establishes that Assumption 1 is irrelevant if a single informed trader trades multiple units of the asset.

Lemma 12. *Under the fee structure in Assumption 1, the informed trader never splits her order. Therefore, the path dependence due to Assumption 1 is irrelevant to the results.*

Proof. Assume that $\tilde{v} = 1+\sigma$, and a single informed trader is buying x_A units of the asset in the AMM.[41] She may buy x_A units of the asset by splitting her orders into $x_{A,1}$ and $x_{A,2}$ units, so that $x_A = x_{A,1} + x_{A,2}$. Equation (B-18) suggests that her payment for the first order is

$$g_1(x_{A,1}, X) = \frac{1}{1-\tau}\frac{Xx_{A,1}}{X-x_{A,1}},$$

while that for the second order is

$$g_2(x_{A,1}, x_{A,2}, X) = \frac{1}{1-\tau}\frac{x_{A,2}}{X-x_A}\left(X + \frac{1}{1-\tau}\frac{Xx_{A,1}}{X-x_{A,1}}\right).$$

[41] The symmetric argument holds when $\tilde{v} = 1-\sigma$.

Therefore, the total payment is

$$\begin{aligned} G(x_{A,1}, x_{A,2}, X) &\equiv g_1(x_{A,1}, X) + g_2(x_{A,1}, x_{A,2}, X) \\ &= \frac{1}{1-\tau}\left[\frac{Xx_{A,1}}{X - x_{A,1}} + \frac{x_A - x_{A,1}}{X - x_A}\left(X + \frac{1}{1-\tau}\frac{Xx_{A,1}}{X - x_{A,1}}\right)\right]. \end{aligned}$$

In contrast, if she trades x_A units all at once, the cost is

$$g_1(x_A, X) = \frac{1}{1-\tau}\frac{Xx_A}{X - x_A}.$$

As long as $\tau > 0$, it holds that $G(x_{A,1}, x_{A,2}, X) > g_1(x_A, X)$, meaning that splitting her order is suboptimal. □

B.3 Concentrated Liquidity in Uniswap v3

When liquidity provider i takes strategy $s(l, u)$, her liquidity parameter is determined by

$$k(l,u) = \begin{cases} w\left(\frac{1}{\sqrt{p_l}} - \frac{1}{\sqrt{p_u}}\right)^{-1} & \text{if } p_0 < p_l, \\ w\left(2 - \sqrt{p_l} - \frac{1}{\sqrt{p_u}}\right) & \text{if } p_0 \in [p_l, p_u], \\ w\left(\sqrt{p_u} - \sqrt{p_l}\right) & \text{if } p_0 > p_u. \end{cases} \tag{B-32}$$

Also, according to the bonding curve in (37), the marginal price of the asset is given by

$$p = -\frac{dC}{dX}\Big|_{(c,x)} = \left(\frac{K_j}{x + \frac{K_j}{\sqrt{p_{j+1}}}}\right)^2. \tag{B-33}$$

If incoming orders buy the asset, for example, they shift the liquidity pool downward along the curve, as described in Figure 7, thereby increasing the marginal price. When the pool hits the intercept with c-axis, all the assets in $s(j, j+1)$ is consumed (i.e., $x \to 0$), and the marginal price becomes $p = p_{j+1}$ following equation (B-33). If the price increases even more, then we move to the next range $s(j+1, j+2)$, in which the same algorithm determines the price. The symmetric argument applies to sell orders that lower the price.

B.3.1 Proof of Propositions 4 and 5

Firstly, we show that the marginal price $p(\delta, \boldsymbol{K})$ satisfies the following properties.

Lemma 13. *(i)* $\frac{\partial p(\delta,\boldsymbol{K})}{\partial \delta} > 0$.
(ii) $\frac{\partial p(\delta,\boldsymbol{K})}{\partial K_j} < 0$ *if and only if* $\delta > 0$ *for* $j \in [l, u]$, *and* $\frac{\partial p(\delta,\boldsymbol{K})}{\partial K_j} = 0$ *for* $j \notin [l, u]$.

Proof. Condition (i) is straightforward from the bonding curve and the definition of p in (B-33), and thus we focus on condition (ii). Consider a target price $\bar{p} > p_0$ such that $\bar{p} \in s(m, m+1)$. Starting with p_0, we derive the quantity of liquidity required to be traded to achieve $p_0 \to \bar{p}$. Firstly, all liquidity in ranges up to $s(m-1, m)$ must be consumed, meaning that the total amount of the asset bought throughout these ranges is

$$\Delta_{0,m-1} \equiv \sum_{j=0}^{m-1} \int k_{i,j} \left(\frac{1}{\sqrt{p_j}} - \frac{1}{\sqrt{p_{j+1}}} \right) di = \sum_{j=0}^{m-1} K_j \left(\frac{1}{\sqrt{p_j}} - \frac{1}{\sqrt{p_{j+1}}} \right).$$

In range $s(m, m+1)$, the liquidity position (x, y) that achieves $\bar{p}$ is determined by solving the marginal price equation (B-33):

$$\sqrt{\bar{p}} = \frac{K_m}{y + \frac{K_m}{\sqrt{p_{m+1}}}} \Rightarrow y = K_m \left(\frac{1}{\sqrt{\bar{p}}} - \frac{1}{\sqrt{p_{m+1}}} \right).$$

This implies that the amount of the asset bought in this range is

$$\Delta_{m,m+1} \equiv y_m - K_m \left(\frac{1}{\sqrt{\bar{p}}} - \frac{1}{\sqrt{p_{m+1}}} \right) = K_m \left(\frac{1}{\sqrt{p_m}} - \frac{1}{\sqrt{\bar{p}}} \right).$$

Therefore, the price moves to $\bar{p}$ if an incoming order buys the following amount of the asset in total:

$$\begin{aligned} \delta(\bar{p}, \boldsymbol{K}) &= \Delta_{0,m-1} + \Delta_{m,m+1} \\ &= K_m \left(\frac{1}{\sqrt{p_m}} - \frac{1}{\sqrt{\bar{p}}} \right) + \sum_{j=0}^{m-1} K_j \left(\frac{1}{\sqrt{p_j}} - \frac{1}{\sqrt{p_{j+1}}} \right). \end{aligned}$$

This can be seen as the inverse price function. It indicates that, for $0 \leq j \leq m$,

$$\frac{\partial \delta(\bar{p}, \boldsymbol{K})}{\partial K_j} > 0,$$

while $\frac{\partial \delta(\bar{p}, \boldsymbol{K})}{\partial K_j} = 0$ for $j > m$. Therefore, increases in liquidity makes it harder for a given order to move the price, as it must buy a larger quantity of the asset. Hence, it shifts down the marginal price curve. Note that the symmetric argument holds for $j < 0$, leading to condition (ii). Also, even if liquidity improves in $s(j, j+1)$ such that $1 + a(\beta, \alpha) < p_j \leq 1 + \sigma$, it does not affect the equilibrium because this price is never reached. □

Secondly, with Lemma 13, Proposition 4 is straightforward.

Proof of Proposition 4. In the proof of Proposition 1 in Appendix C.3, replacing the partial derivatives of variables with respect to X with those with respect to K_j leads to the result. □

Lastly, Proposition 5 follows from the proof below.

Proof. With (l, u) defined in Proposition 4, suppose that the optimal strategy, s^*, leads to $V^* = \max_{l,u} V(l, u)$, yet involves $K_j = 0$ for some $j \in [l, u]$. Then, the profit function and the weighted average expression indicate that $V(j, j+1) > V^*$ because $\lim_{K_j=0} V(j, j+1) = \infty$, contradicting to the optimality of s^*. Therefore, within $s(l, u)$, all price ranges must have $K > 0$. For this liquidity distribution to be an equilibrium, the weighted average expression in (40) implies that all price ranges $s(j, j+1) \subset s(l, u)$ generate the same profit level, while $s(i, i+1) \subsetneq s(l, u)$ leads to zero profit (by construction, the price never moves through these ranges). Therefore, $\boldsymbol{K}$ in the equilibrium must be such that $V(j, j+1) = V(i, i+1)$ for all $i, j \in [l, u]$, implying that $K_j > 0$ in this range. □

C Proof

We assume the following technical conditions to guarantee the uniqueness of the equilibrium price and rule out its abnormal behavior. All conditions lead to the pricing function in Lemma 1 which provides intuition for the regularity conditions.

Condition 1 (Technical conditions)**.** *The CFMM function f with initial liquidity pools (C, X) satisfies the following conditions.*
(i) f is differentiable, $\frac{\partial^2 f}{\partial c \partial x}$ exists, and $f_c(c, x)$ and $f_x(c, x)$ are positive for all $c, x > 0$;
(ii) If (c, x) is on the bonding curve of function f, there exists $h : \mathbb{R}_{++} \to \mathbb{R}_{++}$ such that $c = h(x; C, X)$ and h is decreasing in x;
(iii) $\{(c, x) | f(c, x) > a\}$ is a strictly convex set for all $a > 0$;
(iv) $\frac{f_x(h(X-\delta;C,X),X-\delta)}{f_c(h(X-\delta;C,X),X-\delta)}$ is decreasing in X if and only if $\delta > 0$, convex in δ, and differentiable with respect to δ and X. Moreover, $\partial \frac{f_x(h(X-\delta;C,X),X-\delta)}{f_c(h(X-\delta;C,X),X-\delta)} / \partial X$ is decreasing in δ; and
(v) $a(\delta, X) \equiv |\frac{f_x(h(X-\delta;C,X),X-\delta)}{f_c(h(X-\delta;C,X),X-\delta)} - 1|$ is log-sub-modular in (X, δ).

C.1 Proof of Lemma 2

Proof. The equilibrium condition for noise traders is

$$\bar{\tau}\bar{\gamma}\alpha = \frac{\sigma}{2} \sum_{i=buy,sell} \frac{(1-\beta_i)\eta}{(1-\beta_i)\eta + (1-\eta)(1-\alpha)z} = H(\alpha, \beta_{buy}, \beta_{sell}). \tag{C-34}$$

We have $\frac{\partial H}{\partial \alpha} > 0$, $\frac{\partial^2 H}{\partial \alpha^2} > 0$, $H(0) = \frac{\sigma}{2} \sum_{i=buy,sell} \frac{(1-\beta_i)\eta}{(1-\beta_i)\eta+(1-\eta)z} > 0$, and $H(1) = \sigma$. Condition (17) implies $\bar{\tau}\bar{\gamma} > \sigma$ and guarantees a unique solution to the above equation in $\alpha \in (0, 1)$. It is straightforward to see that the solution is decreasing in β_i, as the partial derivative of the RHS with respect to β_i is negative. □

C.2 Proof of Lemma 3

Proof. β^*_{buy} solves

$$1 + \sigma \frac{(1-\beta_{buy})\eta}{(1-\beta_{buy})\eta + (1-\eta)(1-\alpha)z} = \tilde{p}(\beta_{buy})$$

where $\tilde{p} = \frac{p}{1-\tau}$. This can be rewritten as

$$\frac{\eta}{(1-\eta)z} \frac{(1-\beta_{buy})(1+\sigma-\tilde{p}(\beta_{buy}))}{\tilde{p}(\beta_{buy}) - 1} = 1 - \alpha.$$

The LHS is decreasing in β_{buy}. Thus, we have a unique solution, which is increasing in α. Analogously, β^*_{sell} solves

$$\frac{\eta}{(1-\eta)z}\frac{(1-\beta_{sell})(\hat{p}(-\beta_{sell})-1+\sigma)}{1-\hat{p}(-\beta_{sell})}=1-\alpha$$

with $\hat{p}=\frac{p}{1+\tau}$. The LHS is decreasing in β_{sell}, leading to a unique solution that is increasing in α.

Note that we have $p(\beta)-1>1-p(-\beta)$ and $\frac{1}{1-\tau}>\frac{1}{1+\tau}$. Therefore,

$$\frac{(1-\beta)(\hat{p}(-\beta)-1+\sigma)}{1-\hat{p}(-\beta)}>\frac{(1-\beta)(1+\sigma-\tilde{p}(\beta))}{\tilde{p}(\beta)-1},$$

leading to $\beta^*_{sell}>\beta^*_{buy}$. □

C.3 Proof of Proposition 1

By using (C-34), it holds that

$$\frac{\bar{\tau}}{\sigma}\bar{\gamma}\frac{\partial\alpha}{\partial X}=\frac{z(1-\eta)\eta}{2}\sum_{i=buy,sell}\frac{-(1-\alpha)\frac{\partial\beta_i}{\partial X}+(1-\beta_i)\frac{\partial\alpha}{\partial X}}{L_i} \tag{C-35}$$

where we denote $L_i=((1-\beta_i)\eta+(1-\eta)(1-\alpha)z)^2$. We rewrite this equation as

$$\left[\frac{\bar{\tau}}{\sigma}\bar{\gamma}-\frac{z(1-\eta)\eta}{2}\sum\frac{1-\beta_i}{L_i}\right]\frac{\partial\alpha}{\partial X}=-\frac{z(1-\eta)\eta}{2}(1-\alpha)\sum\frac{\frac{\partial\beta_i}{\partial X}}{L_i}. \tag{C-36}$$

Also, from (19) and (20),

$$\frac{\frac{\partial\beta_{buy}}{\partial X}p_\beta(\beta_{buy},X)+p_X(\beta_{buy},X)}{1-\tau}=\sigma z(1-\eta)\eta\frac{-(1-\alpha)\frac{\partial\beta_{buy}}{\partial X}+(1-\beta_{buy})\frac{\partial\alpha}{\partial X}}{L_{buy}},$$

$$\frac{-\frac{\partial\beta_{sell}}{\partial X}p_\beta(-\beta_{sell},X)+p_X(-\beta_{sell},X)}{1+\tau}=-\sigma z(1-\eta)\eta\frac{-(1-\alpha)\frac{\partial\beta_{sell}}{\partial X}+(1-\beta_{sell})\frac{\partial\alpha}{\partial X}}{L_{sell}},$$

where p_β and p_X denote the partial derivative of p with respect to β and X, respectively. These equations can be expressed as

$$\frac{\partial \beta_{buy}}{\partial X} = -\frac{\frac{L_{buy} p_X(\beta_{buy}, X)}{(1-\tau)\sigma z(1-\eta)\eta} - (1-\beta_{buy})\frac{\partial \alpha}{\partial X}}{(1-\alpha) + \frac{L_{buy} p_\beta(\beta_{buy}, X)}{(1-\tau)\sigma z(1-\eta)\eta}},$$

$$\frac{\partial \beta_{sell}}{\partial X} = \frac{\frac{L_{sell} p_X(-\beta_{sell}, X)}{(1+\tau)\sigma z(1-\eta)\eta} + (1-\beta_{sell})\frac{\partial \alpha}{\partial X}}{(1-\alpha) + \frac{L_{sell} p_\beta(-\beta_{sell}, X)}{(1+\tau)\sigma z(1-\eta)\eta}}.$$

Therefore,

$$\begin{aligned}
\sum \frac{\frac{\partial \beta_i}{\partial X}}{L_i} &= -\frac{1}{L_{buy}} \frac{\frac{L_{buy} p_X(\beta_{buy}, X)}{(1-\tau)\sigma z(1-\eta)\eta} - (1-\beta_{buy})\frac{\partial \alpha}{\partial X}}{(1-\alpha) + \frac{L_{buy} p_\beta(\beta_{buy}, X)}{(1-\tau)\sigma z(1-\eta)\eta}} + \frac{1}{L_{sell}} \frac{\frac{L_{sell} p_X(-\beta_{sell}, X)}{(1+\tau)\sigma z(1-\eta)\eta} + (1-\beta_{sell})\frac{\partial \alpha}{\partial X}}{(1-\alpha) + \frac{L_{sell} p_\beta(-\beta_{sell}, X)}{(1+\tau)\sigma z(1-\eta)\eta}} \\
&= -\frac{\frac{p_X(\beta_{buy}, X)}{(1-\tau)\sigma z(1-\eta)\eta}}{(1-\alpha) + \frac{L_{buy} p_\beta(\beta_{buy}, X)}{(1-\tau)\sigma z(1-\eta)\eta}} + \frac{\frac{p_X(-\beta_{sell}, X)}{(1+\tau)\sigma z(1-\eta)\eta}}{(1-\alpha) + \frac{L_{sell} p_\beta(-\beta_{sell}, X)}{(1+\tau)\sigma z(1-\eta)\eta}} \\
&+ \left(\frac{1}{L_{buy}} \frac{(1-\beta_{buy})}{(1-\alpha) + \frac{L_{buy} p_\beta(\beta_{buy}, X)}{(1-\tau)\sigma z(1-\eta)\eta}} + \frac{1}{L_{sell}} \frac{(1-\beta_{sell})}{(1-\alpha) + \frac{L_{sell} p_\beta(-\beta_{sell}, X)}{(1+\tau)\sigma z(1-\eta)\eta}} \right) \frac{\partial \alpha}{\partial X}.
\end{aligned}$$

Plugging this back into (C-36),

$$Q\frac{\partial \alpha}{\partial X} = -\frac{z(1-\eta)\eta}{2}(1-\alpha)\left[-\frac{\frac{p_X(\beta_{buy}, X)}{(1-\tau)\sigma z(1-\eta)\eta}}{(1-\alpha) + \frac{L_{buy} p_\beta(\beta_{buy}, X)}{(1-\tau)\sigma z(1-\eta)\eta}} + \frac{\frac{p_X(-\beta_{sell}, X)}{(1+\tau)\sigma z(1-\eta)\eta}}{(1-\alpha) + \frac{L_{sell} p_\beta(-\beta_{sell}, X)}{(1+\tau)\sigma z(1-\eta)\eta}} \right] \tag{C-37}$$

where Q is

$$
\begin{aligned}
Q &= \frac{\bar{\tau}}{\sigma}\bar{\gamma} + \frac{z(1-\eta)\eta}{2}\frac{1}{L_{buy}}\frac{(1-\alpha)(1-\beta_{buy})}{(1-\alpha)+\frac{L_{buy}p_\beta(\beta_{buy},X)}{(1-\tau)\sigma z(1-\eta)\eta}} \\
&+ \frac{z(1-\eta)\eta}{2}\left(\frac{1}{L_{sell}}\frac{(1-\alpha)(1-\beta_{sell})}{(1-\alpha)+\frac{L_{sell}p_\beta(-\beta_{sell},X)}{(1+\tau)\sigma z(1-\eta)\eta}} - \sum_{i=buy,sell}\frac{1-\beta_i}{L_i}\right) \\
&= \frac{\bar{\tau}}{\sigma}\bar{\gamma} - \frac{1}{2}\frac{1}{1-\alpha}\left(\frac{(1-\beta_{buy})\frac{p_\beta(\beta_{buy},X)}{(1-\tau)\sigma}}{1+\frac{L_{buy}}{z(1-\eta)\eta(1-\alpha)}\frac{p_\beta(\beta_{buy},X)}{(1-\tau)\sigma}} + \frac{(1-\beta_{sell})\frac{p_\beta(-\beta_{sell},X)}{(1+\tau)\sigma}}{1+\frac{L_{sell}}{z(1-\eta)\eta(1-\alpha)}\frac{p_\beta(-\beta_{sell},X)}{(1+\tau)\sigma}}\right) \\
&> \frac{\bar{\tau}}{\sigma}\bar{\gamma} - 1.
\end{aligned}
$$

The last inequality comes from

$$
1-\beta_i < \frac{L_i}{z(1-\eta)\eta(1-\alpha)}(1-\alpha).
$$

To prove this inequality, denote $\tilde{p}_{buy} = \frac{p(\beta_{buy},X)}{1-\tau}$ and observe that $L_{buy} = \left(\frac{\sigma}{\tilde{p}_{buy}-1}\right)^2(1-\beta_{buy})^2\eta^2$ and $(1-\beta_{buy})z(1-\eta)\eta = \frac{1}{1-\alpha}\frac{1+\sigma-\tilde{p}_{buy}}{\tilde{p}_{buy}-1}(1-\beta_{buy})^2\eta^2$. Therefore, the following condition holds:

$$
\begin{aligned}
L_{buy} - (1-\beta_{buy})z(1-\eta)\eta &= \frac{(1-\beta_{buy})^2\eta^2(\tilde{p}_{buy}-1)^2}{1-\alpha} \\
&\quad \times \left[(1-\alpha)\sigma^2 - (\tilde{p}_{buy}-1)(1+\sigma-\tilde{p}_{buy})\right].
\end{aligned}
$$

Thus, it suffices to show $(1-\alpha)\sigma^2 > (\tilde{p}_{buy}-1)(1+\sigma-\tilde{p}_{buy})$. Since $\alpha = \frac{1}{2\bar{\gamma}\bar{\tau}}(\tilde{p}_{buy}-\tilde{p}_{sell})$, it holds that

$$\begin{aligned}
(1-\alpha)\sigma^2 - (\tilde{p}_{buy}-1)(1+\sigma-\tilde{p}_{buy}) &= (1-\frac{1}{2\bar{\gamma}\bar{\tau}}(\tilde{p}_{buy}-\tilde{p}_{sell}))\sigma^2 - (\tilde{p}_{buy}-1)(1+\sigma-\tilde{p}_{buy}) \\
&> \tilde{p}_{buy}^2 - \left(2+\sigma+\frac{\sigma^2}{2\bar{\gamma}\bar{\tau}}\right)\tilde{p}_{buy} + \sigma^2+\sigma+1+\frac{\sigma^2(2-\tilde{p}_{buy})}{2\bar{\gamma}\bar{\tau}} \\
&\geq \frac{\sigma^2}{4}\left(3+\frac{\sigma}{\bar{\gamma}\bar{\tau}}\right)\left(1-\frac{\sigma}{\bar{\gamma}\bar{\tau}}\right) \\
&> 0.
\end{aligned}$$

Therefore, we have established $1-\beta_{buy} < \frac{(1-\alpha)L_{buy}}{z(1-\eta)\eta(1-\alpha)}$. Note that $\frac{\bar{\tau}}{\sigma}\bar{\gamma}-1>0$ is the same condition as (17). As the RHS of (C-37) is negative, we conclude that $\frac{\partial\alpha^*}{\partial X} < 0$. This also implies that the bid-ask spread is decreasing in X.

Next, consider the impact of X on $\beta_{buy}+\beta_{sell}$. Since $\frac{1-\beta_i}{L_i} < \frac{1}{z(1-\eta)\eta}$ and $\frac{\partial\alpha}{\partial X} < 0$, it holds that

$$\sum_{i=buy,sell}\frac{(1-\beta_i)\frac{\partial\alpha}{\partial X}}{L_i} > 2\frac{\frac{\partial\alpha}{\partial X}}{z(1-\eta)\eta}.$$

Equation (C-35) implies

$$\begin{aligned}
\frac{\bar{\tau}\bar{\gamma}}{\sigma}\frac{\partial\alpha}{\partial X} &> \frac{z(1-\eta)\eta}{2}\left[-\sum_{i=buy,sell}\frac{(1-\alpha)\frac{\partial\beta_i}{\partial X}}{L_i} + 2\frac{\frac{\partial\alpha}{\partial X}}{z(1-\eta)\eta}\right] \\
&= -\frac{z(1-\eta)\eta}{2}\sum_{i=buy,sell}\frac{(1-\alpha)\frac{\partial\beta_i}{\partial X}}{L_i} + \frac{\partial\alpha}{\partial X}.
\end{aligned}$$

$$\therefore \frac{z(1-\eta)\eta}{2}\sum_{i=buy,sell}\frac{(1-\alpha)\frac{\partial\beta_i}{\partial X}}{L_i} > -\frac{\partial\alpha}{\partial X}\left(\frac{\bar{\tau}\bar{\gamma}}{\sigma}-1\right).$$

Due to (17), the RHS is positive. Thus, at least one of $\frac{\partial\beta_{buy}}{\partial X}$ and $\frac{\partial\beta_{sell}}{\partial X}$ must be positive. Note that (19) and (20) are rewritten as

$$A\frac{\partial\beta_{buy}}{\partial X} = -\frac{L_{buy}p_X(\beta_{buy},X)}{(1-\tau)\sigma\eta(1-\beta_{buy})} + z(1-\eta)\frac{\partial\alpha}{\partial X} = -\frac{\tilde{p}_X(\beta_{buy},X)}{(\tilde{p}(\beta_{buy})-1)} + \frac{z(1-\eta)}{\sqrt{L_{buy}}}\frac{\partial\alpha}{\partial X},$$

$$B\frac{\partial \beta_{sell}}{\partial X} = \frac{L_{sell} p_X(-\beta_{sell}, X)}{(1+\tau)\sigma\eta(1-\beta_{sell})} + z(1-\eta)\frac{\partial \alpha}{\partial X} = \frac{\hat{p}_X(-\beta_{sell}, X)}{(1-\hat{p}(-\beta_{sell}, X))} + \frac{z(1-\eta)}{\sqrt{L_{sell}}}\frac{\partial \alpha}{\partial X}.$$

where A and B are positive constants. Hence,

$$\begin{aligned} A\frac{\partial \beta_{buy}}{\partial X} - B\frac{\partial \beta_{sell}}{\partial X} &= -\frac{\tilde{p}_X(\beta_{buy}, X)}{\tilde{p}(\beta_{buy}) - 1} - \frac{\hat{p}_X(-\beta_{sell}, X)}{1-\hat{p}(-\beta_{sell}, X)} + \left(\frac{1}{\sqrt{L_{buy}}} - \frac{1}{\sqrt{L_{sell}}}\right) z(1-\eta)\frac{\partial \alpha}{\partial X} \\ &> -\frac{\tilde{p}_X(\beta_{buy}, X)}{\tilde{p}(\beta_{buy}) - 1} - \frac{\hat{p}_X(-\beta_{sell}, X)}{1-\hat{p}(-\beta_{sell}, X)}, \end{aligned}$$

where the last inequality comes from $\beta_{sell} > \beta_{buy}$ and $\frac{\partial \alpha}{\partial X} < 0$. Point (v) in Condition 1 implies that the last expression is positive. Using $\beta_{buy} < \beta_{sell}$, we obtain

$$0 < \frac{\frac{\partial \beta_{buy}}{\partial X}}{L_{buy}} + \frac{\frac{\partial \beta_{sell}}{\partial X}}{L_{sell}} < \frac{\frac{\partial \beta_{buy}}{\partial X} + \frac{\partial \beta_{sell}}{\partial X}}{L_{sell}}.$$

C.4 Proof of Lemmas 5 and 6 and Proposition 2

Proof. Firstly, V_{NT}^{AMM} is monotonically decreasing in X because $\frac{\partial \alpha(X)}{\partial X} < 0$, leading to the result in Lemma 6. Secondly, regarding V_{IT}^{AMM}, it holds that

$$\frac{2}{w}\frac{\partial V_{IT}^{AMM}(X)}{\partial X} = (1-\tau)\left[\frac{p(\beta_{buy}(X), X)}{1-\tau} - (1+\sigma)\right]\frac{\partial \beta_{buy}(X)}{\partial X} \tag{C-38}$$

$$+ (1+\tau)\left[(1-\sigma) - \frac{p(-\beta_{sell}(X), X)}{1+\tau}\right]\frac{\partial \beta_{sell}(X)}{\partial X} \tag{C-39}$$

$$+ \frac{\partial P(\beta_{buy}(X), X)}{\partial X}\beta_{buy}(X) - \frac{\partial P(-\beta_{sell}(X), X)}{\partial X}\beta_{sell}(X).$$

Note that condition (ii) in Lemma 1 leads to

$$\frac{\partial P(\beta_{buy}(X), X)}{\partial X} = \frac{1}{\beta_{buy}}\int_0^{\beta_{buy}} \frac{\partial p(\delta, X)}{\partial X} d\delta < 0,$$

$$\frac{\partial P(-\beta_{sell}(X), X)}{\partial X} = \frac{1}{\beta_{sell}}\int_0^{\beta_{sell}} \frac{\partial p(-\delta, X)}{\partial X} d\delta > 0.$$

Hence, the second line of (C-38) is negative. As for the first line of (C-38), the equilibrium condition implies that

$$\frac{p(\beta_{buy}(X), X)}{1-\tau} - (1+\sigma) = -\sigma \frac{(1-\eta)(1-\alpha)z}{(1-\beta_{buy})\eta + (1-\eta)(1-\alpha)z} < 0,$$

$$(1-\sigma) - \frac{p(-\beta_{sell}(X), X)}{1+\tau} = -\sigma \frac{(1-\eta)(1-\alpha)z}{(1-\beta_{sell})\eta + (1-\eta)(1-\alpha)z} < 0.$$

Moreover, we have established that $A\frac{\partial \beta_{buy}(X)}{\partial X} > B\frac{\partial \beta_{sell}(X)}{\partial X}$ for some positive constants A and B in Appendix C.3. If $\frac{\partial \beta_{sell}(X)}{\partial X} > 0$, then the first line of (C-38) is negative, concluding the proof. Hence, suppose that $\frac{\partial \beta_{sell}(X)}{\partial X} < 0$. In this case, using $\beta_{buy} < \beta_{sell}$, the following inequality holds:

$$\begin{aligned}
\frac{\text{The first line of (C-38)}}{\sigma(1-\eta)(1-\alpha)z} &= -\sum_{j=buy,sell} \sqrt{L_i}\frac{\frac{\partial \beta_j(X)}{\partial X}}{L_j} \\
&< -\sqrt{L_{buy}} \sum_{j=buy,sell} \frac{\frac{\partial \beta_j(X)}{\partial X}}{L_i} \\
&< 0.
\end{aligned}$$

The above inequality completes the proof of Lemma 5.

Finally, the above arguments imply that V_M^{AMM} is monotonically decreasing in X. For the existence and uniqueness of the solution, the behavior of the profit at the limit of $X \to 0$ and $X \to \infty$ must be computed. When $X \to 0$, we have that $\beta_j \to 0$, while $\alpha_0 \equiv \lim_{X\to 0} \alpha(X)$ being the solution to

$$\alpha = \lim_{\beta_j \to 0} \frac{S(\beta, \alpha)}{2\bar{\gamma}\bar{\tau}} = \frac{1}{\bar{\gamma}\bar{\tau}} \frac{\eta}{\eta + (1-\eta)\alpha z}.$$

Hence, given condition (17), $\alpha_0 > 0$. Therefore, we confirm that

$$\lim_{X\to 0} V_M^{AMM} = (1-\eta)w\tau z\alpha_0 > 0.$$

In contrast, at the limit of $X \to \infty$, we have $p(\delta, X) \to 1$. The informed traders' behavior

at this limit, denoted as $\beta_{buy,\infty}$ and $\beta_{sell,\infty}$, is determined by the following conditions:

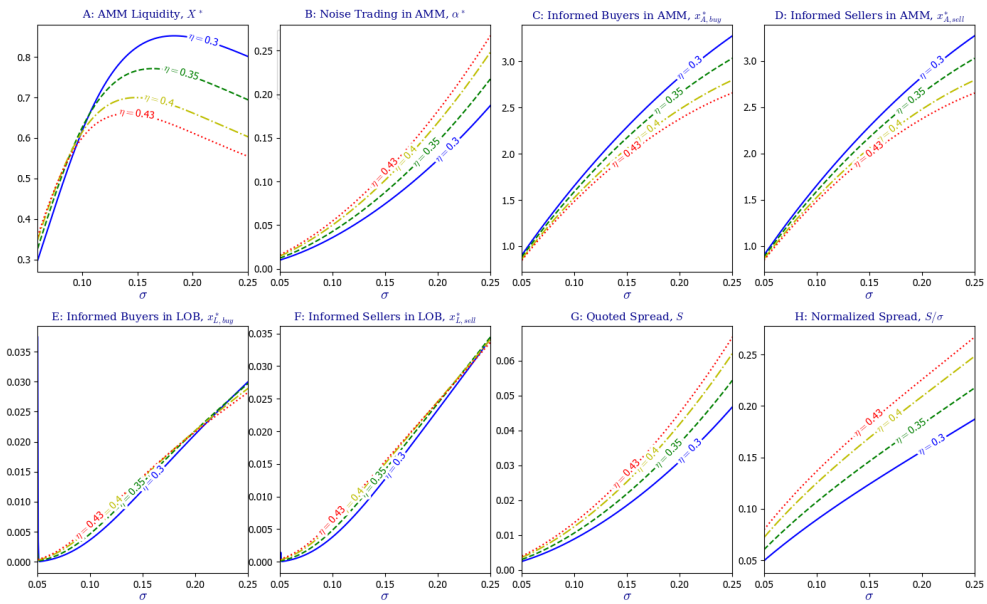
$$\frac{1}{1-\tau} = 1 + \sigma \frac{(1-\beta_{buy})\eta}{(1-\beta_{buy})\eta + (1-\eta)(1-\alpha)z},$$

$$\frac{1}{1+\tau} = 1 - \sigma \frac{(1-\beta_{sell})\eta}{(1-\beta_{sell})\eta + (1-\eta)(1-\alpha)z}.$$

Hence, for all $\alpha \in [0,1)$, it holds that $\beta_{j,\infty} > 0$, leading to $V_{IT}^{AMM} < 0$. In contrast, the above conditions imply that the bid-ask spread becomes $S = 2\bar{\tau}$. Therefore, $\alpha \to 0$ and $V_{NT}^{AMM} \to 0$. Therefore, we conclude that $\lim_{X\to\infty} V_M^{AMM} = V_{IT}^{AMM} < 0$. Together with the monotonicity, there is a unique solution, X^*, to the break-even condition, $V_M^{AMM}(X) = 0$. □

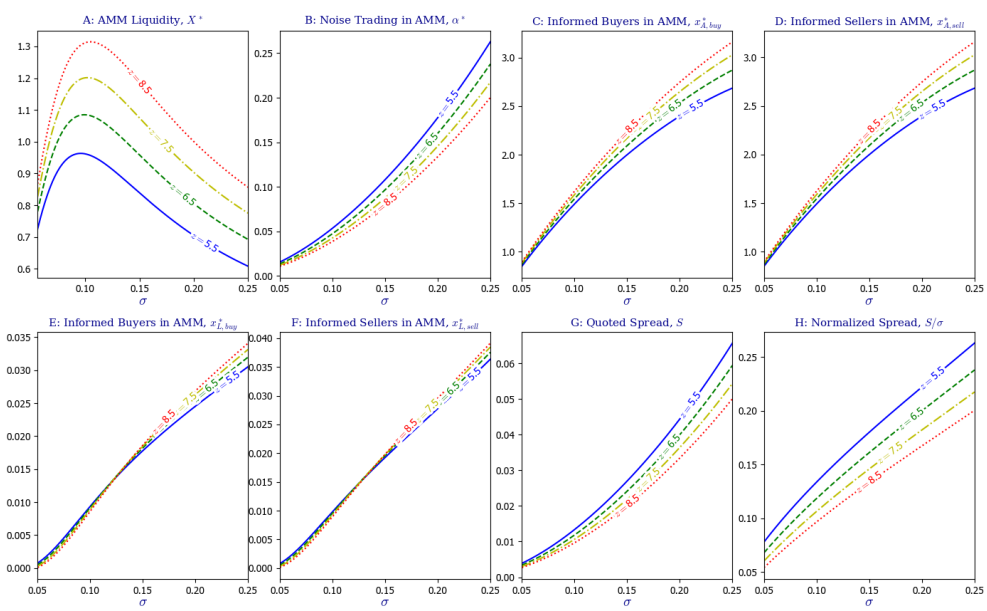
D Additional Figures

Figure 8: Equilibrium in Subsection 5.1 with various η



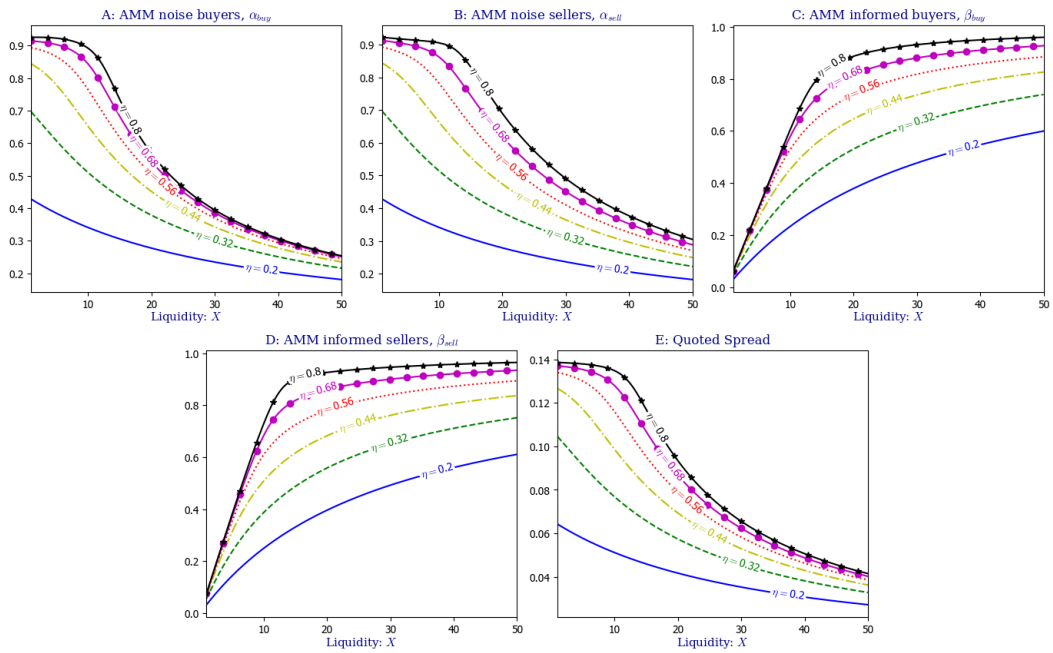
Note: This figure is illustrated by using the following parameter values. $\tau = 0.005$, $z = 7.5$, $\bar{\gamma} = 50$, $\varepsilon = 0.0003$, $c = 0.05$. The parameter values are restricted to satisfy all conditions for the unique coexisting equilibrium.

Figure 9: Equilibrium in Subsection 5.1 with various z



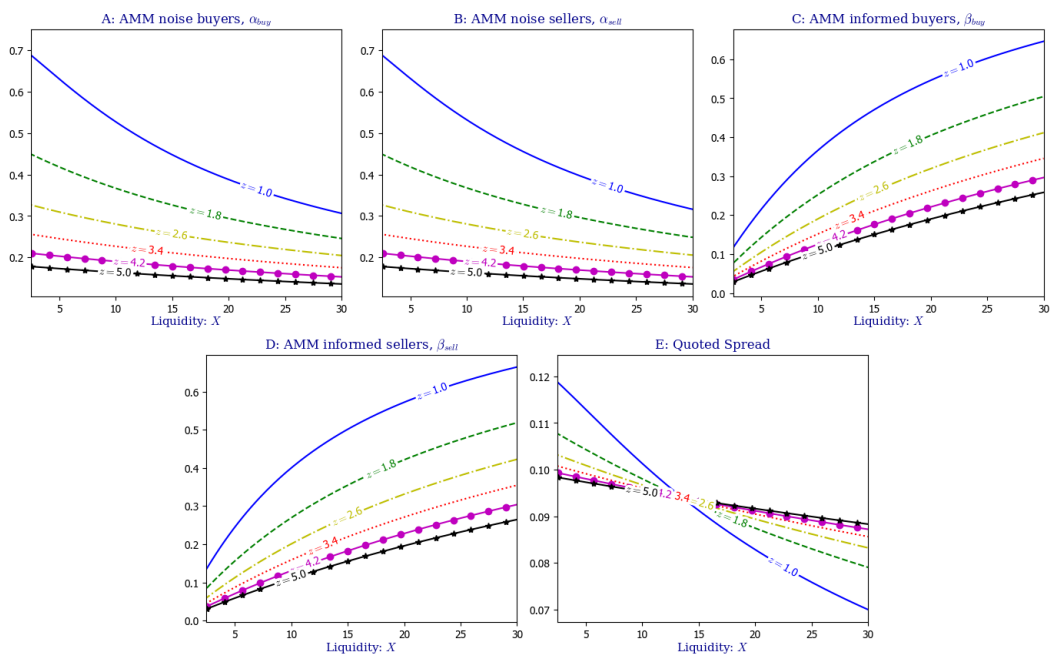
Note: This figure is illustrated by using the following parameter values. $\tau = 0.005$, $\eta = 0.35$, $\bar{\gamma} = 50$, $\varepsilon = 0.0003$, $c = 0.05$. The parameter values are restricted to satisfy all conditions for the unique coexisting equilibrium.

Figure 10: Liquidity Spillover in Subsection 5.2 with various η



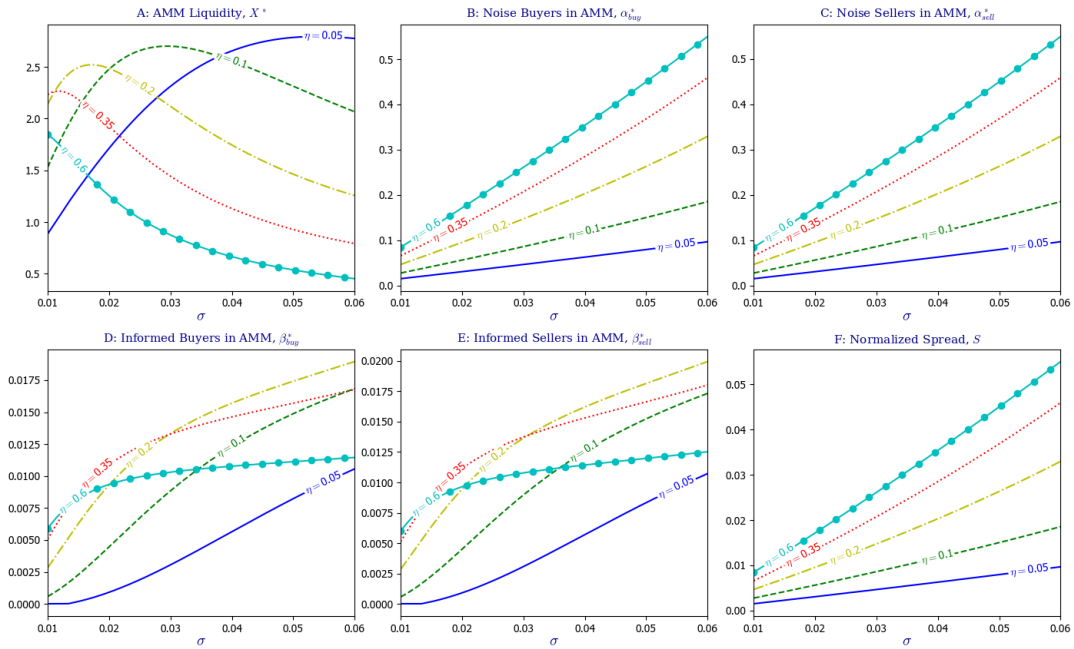
Note: This figure is illustrated by using the following parameter values. $\tau = 0.003$, $z = 0.5$, $\bar{\gamma} = 50$, $\sigma = 0.14$. The parameter values are restricted to satisfy all conditions for the unique coexisting equilibrium.

Figure 11: Liquidity Spillover in Subsection 5.2 with various z



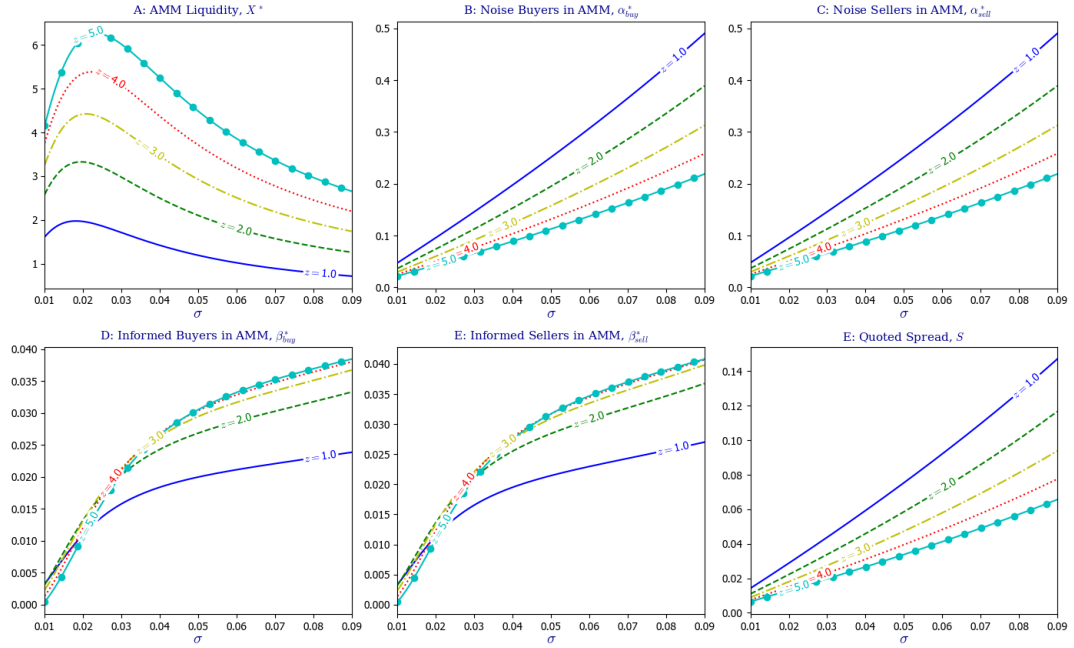
Note: This figure is illustrated by using the following parameter values. $\tau = 0.003$, $\eta = 0.5$, $\bar{\gamma} = 50$, $\sigma = 0.14$. The parameter values are restricted to satisfy all conditions for the unique coexisting equilibrium.

Figure 12: Equilibrium in Subsection 5.2 with various η



Note: This figure is illustrated by using the following parameter values. $\tau = 0.002$, $z = 0.3$, $\bar{\gamma} = 50$, $\varepsilon = 0.00001$. The parameter values are restricted to satisfy all conditions for the unique coexisting equilibrium.

Figure 13: Equilibrium in Subsection 5.2 with various z



Note: This figure is illustrated by using the following parameter values. $\tau = 0.003$, $\eta = 0.7$, $\bar{\gamma} = 50$, $\varepsilon = 0.00005$. The parameter values are restricted to satisfy all conditions for the unique coexisting equilibrium.